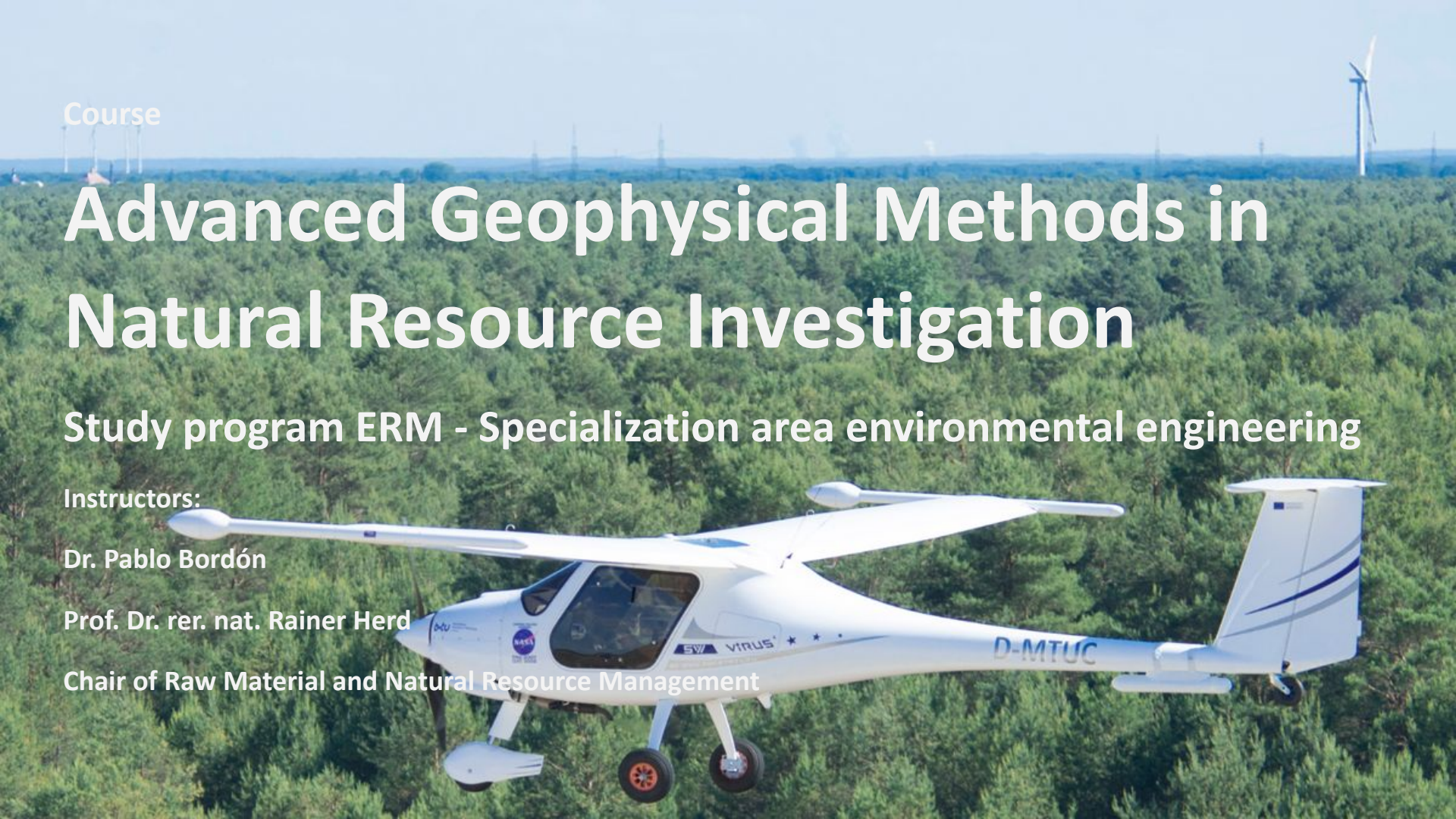
Course

Advanced Geophysical Methods in Natural Resource Investigation

Study program ERM - Specialization area environmental engineering



Instructors:

Dr. Pablo Bordón

Prof. Dr. rer. nat. Rainer Herd

Chair of Raw Material and Natural Resource Management

Course Overview

Duration: 1 semester

Credits: 6

Mandatory prerequisites: Successful completion of the module (*42421*) Natural Resource Investigation.

Limited number of participants: 15

Forms of teaching: Lectures + Field training + Self-study 120 (hours)

Assessment mode for module examination:

- Written exam covering the lecture content (90 minutes)
- Oral presentation of field and lab work (15 minutes)
- Written technical report for the practical training (10-15 pages)

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Techniques that use physical principles (electrical, electromagnetic, magnetic, radiometric) to investigate the subsurface without excavation.

The search for economically valuable materials (e.g. groundwater, minerals) using technical and scientific approaches.

- Civil engineering (e.g. tunnels, foundations)
- Archaeological investigations
- Environmental assessments
- UXO detection

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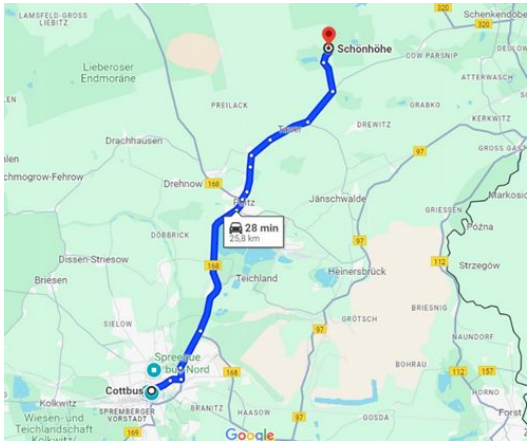
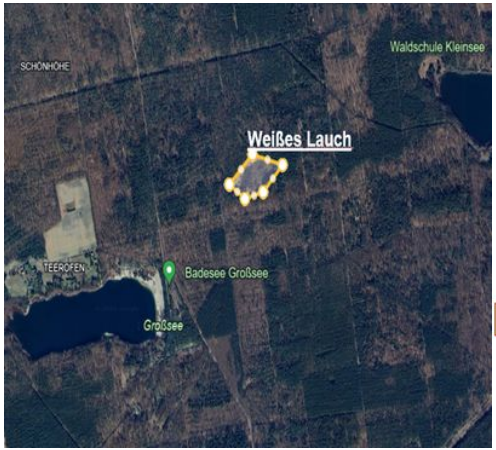
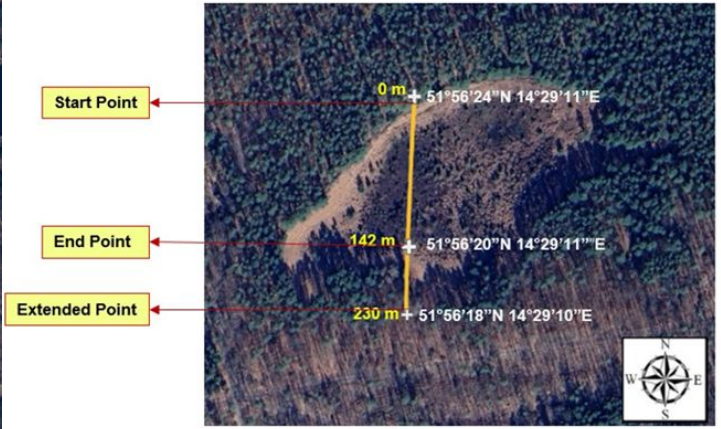
Sounds familiar? …

Weißes Lauch Field Training → Acquire Practical Experience







Setting: Semi-forested area with birch and oak trees, wetlands, and **traces of swamps**

Geological history. Influenced by glacial retreat, hydrological changes, **peat formation**, and climate effects

The aim was to gain field experience using geophysical methods to study subsurface structures and identify raw materials.

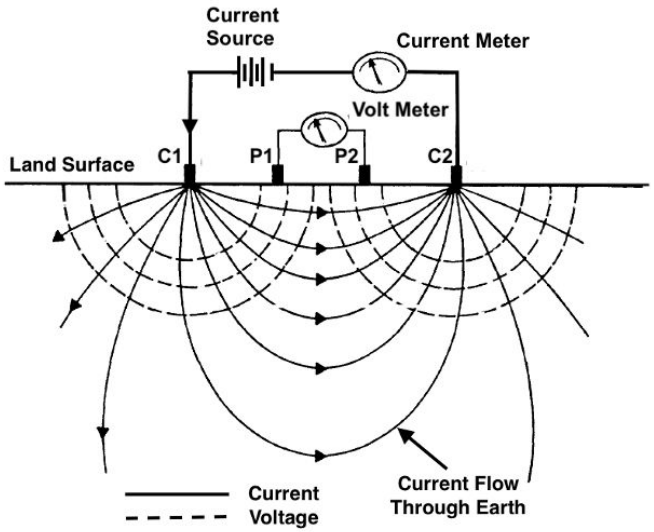
The site contains peat layers, sand, and groundwater. all of them are considered natural resources depending on the context (energy, agriculture, ecology).

Weißes Lauch Field Training → Using ERT

This method allowed you to identify variations in subsurface resistivity linked to:

soil composition, moisture, and geological structures

Weißes Lauch Field Training → ERT Studies → How do we estimate the resistivity?

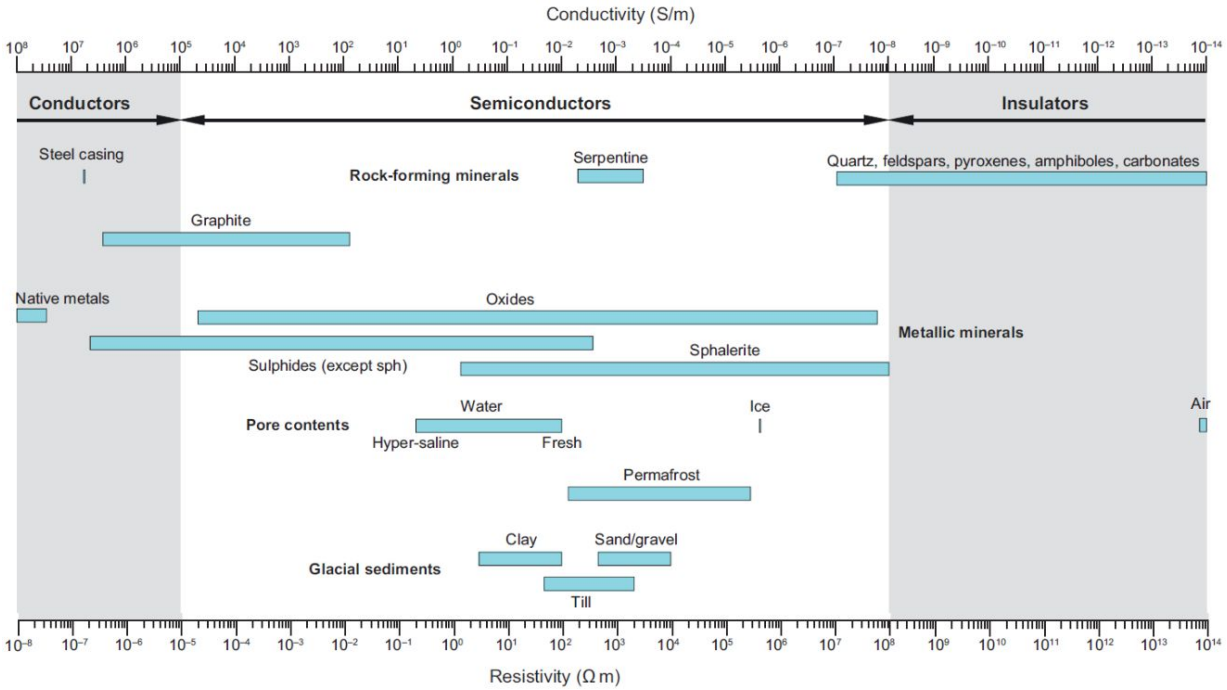


- Determination of the underground situation by measuring of resistivities and comparison with known resistivities of geological materials
- Method is based on *Ohm's Law* $R = \frac{V}{I}$

  R: Resistance (Ω)
  U: Potential (V)
  I: Current (A)
- Uses 4 electrodes, C1 and C2 (current electrodes) and P1 and P2 (potential electrodes)
- Under consideration of the geometric position of the electrodes (K-factor) we obtain the resistivity

Weißes Lauch Field Training → Resistivity of Subsurface Materials

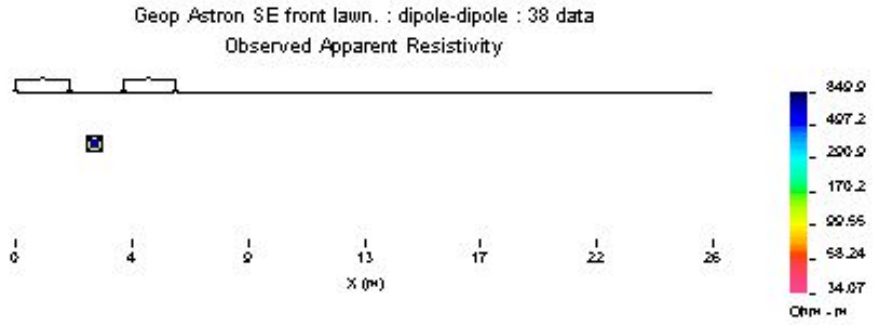
The electrical behaviour of rocks, sediments, and soils is governed by three types of conductivity: **electrolytic**, electronic and dielectric.



Electrolytic conduction.:
This is the dominant mechanism in most geological materials. It occurs through the **movement of ions in pore fluids**.

Resistivity depends on several factors. not just the material itself, but also **moisture content, porosity, salinity, and temperature**.

Weißes Lauch Field Training → ERT Studies → Apparent Resistivity

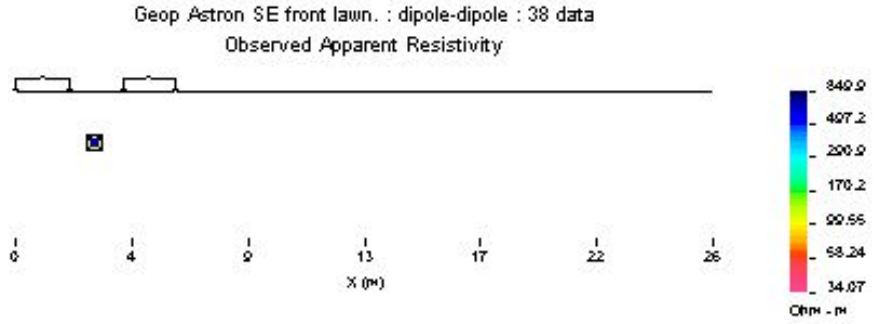


$$\rho_a = k\frac{V}{I}$$

ρa: apparent resistivity (Ωm)
k: Geometric factor, represents the distance between the different electrodes
U: Potential (V)
I: Current (A)

It is based on the assumption that the ground is homogeneous and isotropic (homogeneous half space)

Weißes Lauch Field Training → ERT Studies → Apparent Resistivity



Does this values accurately represents the “true” distribution of subsurface resistivity?

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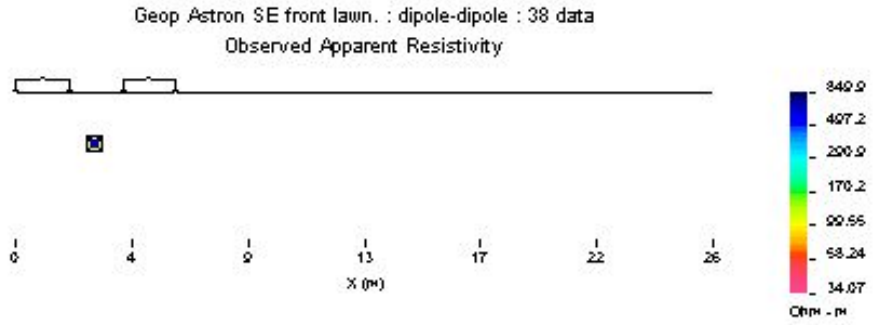

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Does this values accurately represents the “true” distribution of subsurface resistivity?

That apparent resistivity value would correspond to the true resistivity if the subsurface were homogeneous.

It is a fictitious value, it does not represent the actual resistivity of any specific layer

However, if the ground is layered or has heterogeneities, that value is merely an integrated or weighted average response of the actual resistivities present at depth

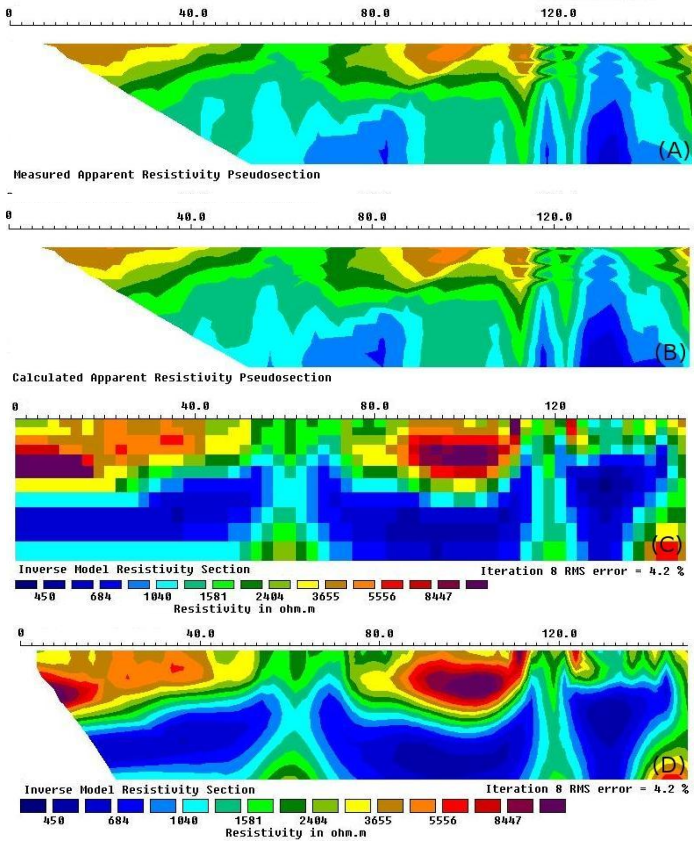
Weißes Lauch Field Training → ERT Studies → Inversion procedure

Measured/Observed resistivity values

Calculated resistivity values

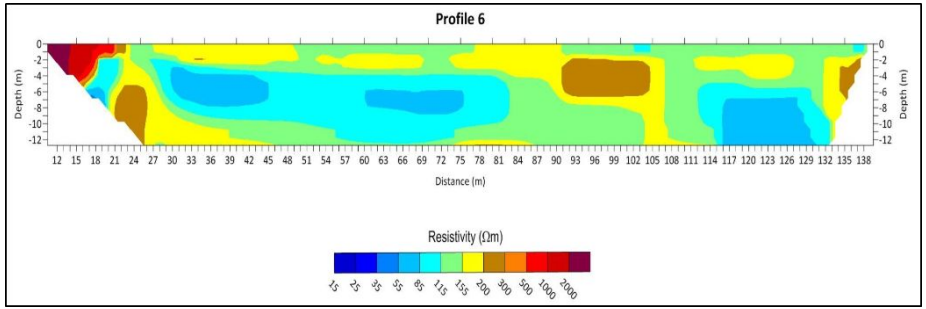
True resistivity values (model blocks)

True resistivity values (contoured section)

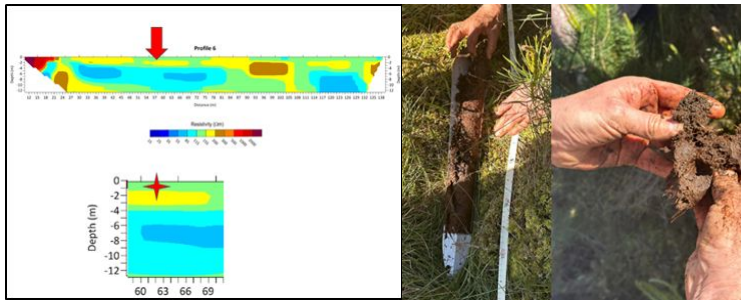


Weißes Lauch Field Training → ERT Studies → Final Resistivity Model

Profiles provided for Interpretation



Drillings at specific locations



With ERT we measure resistivity, but we don't directly see the material underground. **By drilling at specific points, we can observe what's really there.** whether it's sand, peat, clay, or groundwater.

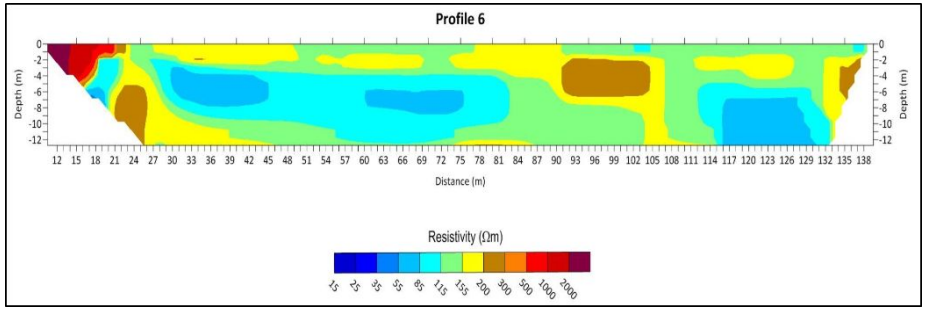
Once we know that a certain resistivity value corresponds to a particular material at one location, **we can extrapolate** that information to other areas in the profile with similar values. assuming the geological context is consistent.

Material	Resistivity in Ωm
Clay (wet)	3 - 30
Peat (wet)	10 - 85
Sand (wet)	100 - 200
Gravel (wet)	200 - 500
Sand (dry)	> 500

Typical resistivities of local geological materials

Weißes Lauch Field Training → ERT Studies → Final Resistivity Model

Profiles provided for Interpretation



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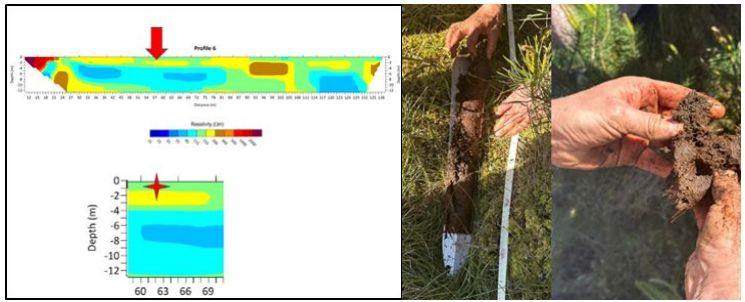

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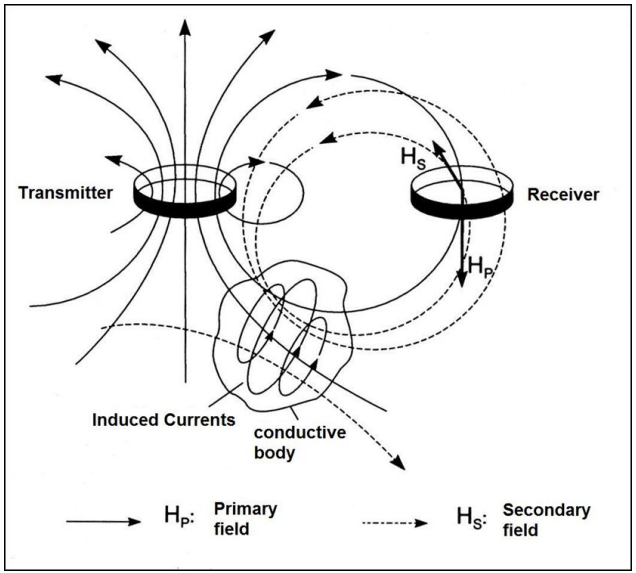
Typical resistivities of local geological materials

So far your experience using this method:

- acquisition
- processing
- interpretation

Weißes Lauch Field Training → EM induction

Electromagnetic (HLEM & VLEM): To measure ground conductivity and infer resistivity indirectly using EM induction



Application of an artificial electromagnetic field, comparison of primary and secondary field, determination of ground conductivity (reciprocal value of resistivity)

$$\frac{H_s}{H_p} \propto \sigma$$

$$\rho = \frac{1}{\sigma}$$

Course Overview

Objective:

- To apply geophysical prospecting methods for the exploration of natural resources and other related applications.

Covering the complete workflow from data acquisition to interpretation.

Contents:

- *Lecture:* Theory about working principles and processing techniques of geophysical prospecting methods.
- *Field training:* Hands-on data acquisition in the field.
- Lab session: Data processing and interpretation.

Focus of the course → Strong practical orientation

Full workflow:

- From data acquisition to interpretation (diff Weißes Lauch) → comprehensive approach

Hands-on training

- Survey planning → selecting appropriate methods based on target, terrain, expected contrasts, and interferences.
- Instrument operation in the field → use of equipment, quality control.

Data processing

- Commercial software → powerful, user friendly
- Open-source tools (Python, QGIS) → flexible toolbox for data management

Geophysical Methods

Electrical Resistivity Tomography

EM (Frequency Domain, Time Domain)

Geophysical Methods

Seismic

Gamma-Ray spectrometry (K, Th, U)

Electrical Resistivity Tomography

Magnetotellurics (MT)

Self Potential (SP)

Ground Penetrating Radar (GPR)

VLF (very low frequency)

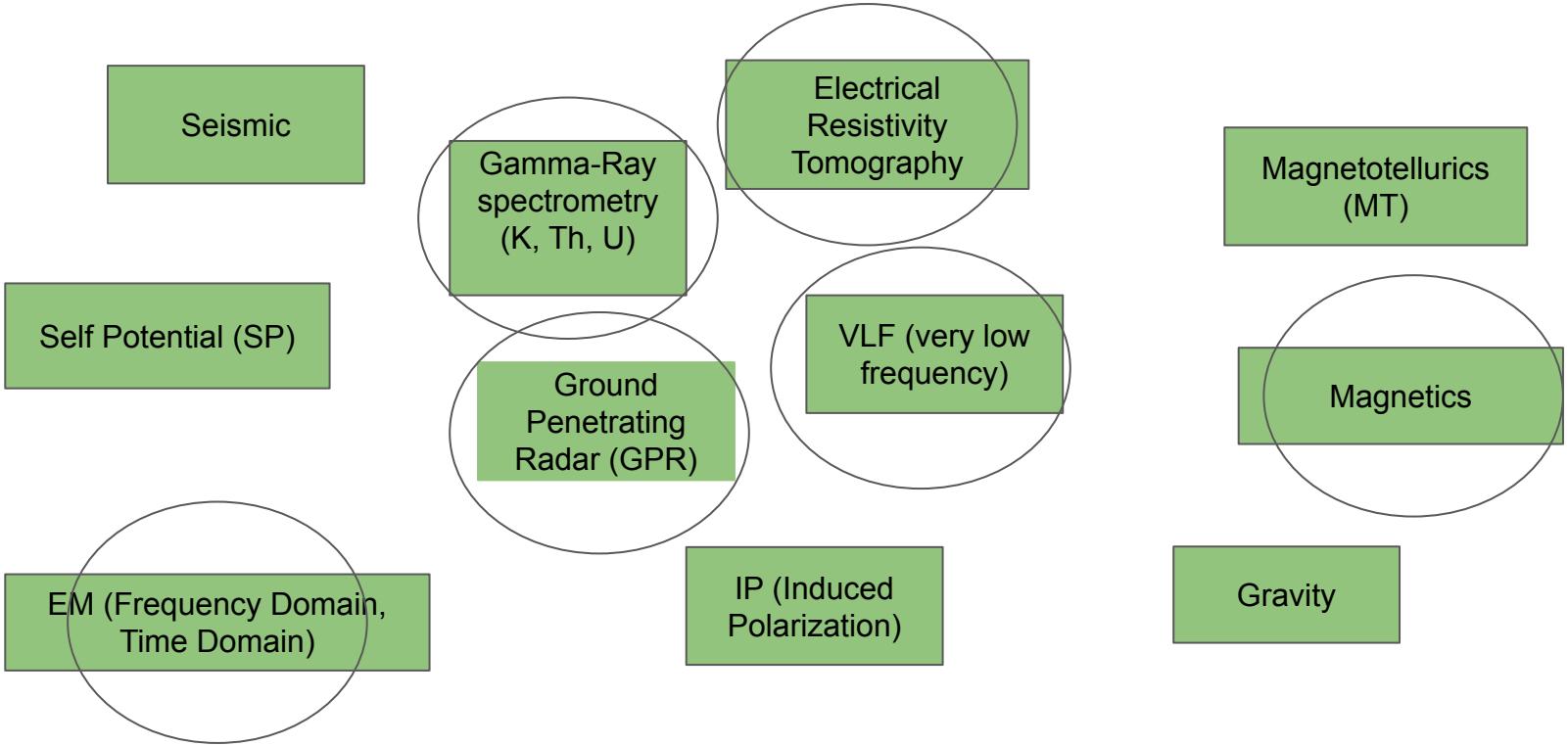
Magnetics

EM (Frequency Domain, Time Domain)

IP (Induced Polarization)

Gravity

Geophysical Methods



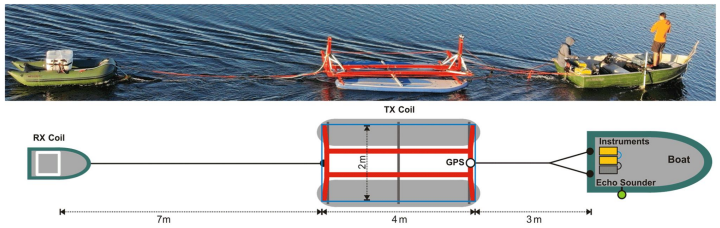
Classification based on deployment method

Geophysical surveys can be conducted:

- from the air
- on the ground
- in marine environments.

The choice depends on factors such a:

- scale
- resolution
- accessibility
- and target depth.



Classification based on deployment method

Airborne methods

- Cover large areas efficiently and rapidly.
- Ideal for regional mapping or inaccessible terrain.
- Common techniques: magnetometry, radiometry, EM, gravimetry.

Ground-based methods

- Offer higher resolution and greater data density.
- Used for detailed site characterization or follow-up investigations.
- Common techniques: ERT, GPR, seismic, magnetics, EM.

Marine-based methods

- Applied in offshore or coastal environments.
- Used for seafloor mapping, resource exploration, and environmental studies.
- Common techniques: marine seismic, magnetics, EM, gravity.

Airborne surveys are used in the early exploration stages,

Ground or marine surveys for detailed investigations.

Methods you will learn in this course

Geophysical methods covered

Ground-based

- **ERT** (subsurface resistivity) → groundwater, structural studies → **Focusing on processing-modelling**
- **EM induction** (electrical conductivity) → environmental and mineral applications → **Focusing on processing-modelling**
- **GPR** (dielectric permittivity and signal reflections) → shallow features like pipes or archaeological remains → **Full workflow : acquisition - processing - interpretation**
- **Magnetometry** (measures magnetic field intensity) → detection of ferrous objects, Unexploded Ordnance (UXO) Detection. → **Full workflow : acquisition - processing - interpretation**

Methods you will learn in this course

Geophysical methods covered

Airborne- based

- **Magnetometry** (measures magnetic field intensity) → regional geological mapping, detection of faults, dikes, mineral exploration (e.g. Fe deposits associated with magnetic minerals suchs a magnetite). → **Acquisition - Processing - Interpretation.**
- **Radiometry** (measures gamma radiation from K, U, Th) → mapping of surface lithology, mineral exploration (e.g, uranium, rare earths, potassic alteration) → **Acquisition - Processing - Interpretation.**
- **VLF-EM** (passive EM response to natural VLF transmitters) → detection of shallow conductors, fracture zones, hydrogeological studies. → **Processing - Interpretation.**

Preliminary schedule (in progress)

# Lecture num	Date	Topic	Accions	practical training
1	16/10	Presentation	methods & use cases & physics - projects of the chair	slide show them the programs
2	23/10	Geophysical prospecting methods	Theory	
3	30/10	GPR 1	Aquisition	
4	06/11	GPR 2	Processing GPR	
5	13/11	GIS	QGIS training	
6	20/11	Magnetometer 1	Theory, Aquisition	
7	27/11	Magnetometer 2	Aquisition, Interpretation	
8	04/12	ERT 1	Theory	
9	11/12	ERT 2	Processing	
10	18/12	About different methods	explain a paper or a project for next week	
11	25/12	Holiday		
12	01/01	Holiday		
13	08/01	Paper explanation		
14	15/01	Airborne data 1	Processing theory	
15	22/01	Airborne data 2	Theory - specific detailed Study cases - papers - Processing theory	
16	29/01	Airborne data 3		
17	05/02	Airborne data 4		
18	12/02	Review	Review topis for the exam	
19	19/02	Exam 1		
20	26/02	Exam 2		

Upcoming classes… 30.10 & 6.11

Goals for the next classes: we start with our first method:

GPR (ground penetrating radar)

Theory

- How GPR works (electromagnetic pulses, reflections)
- Physical principles and signal behavior in different materials
- Applications: shallow structures, archaeology, utilities
- Data processing steps

Acquisition

- Field setup and data collection procedures
- Equipment handling and survey design basics

Lab Session

- Practice with GPR software
- Basic interpretation of profiles

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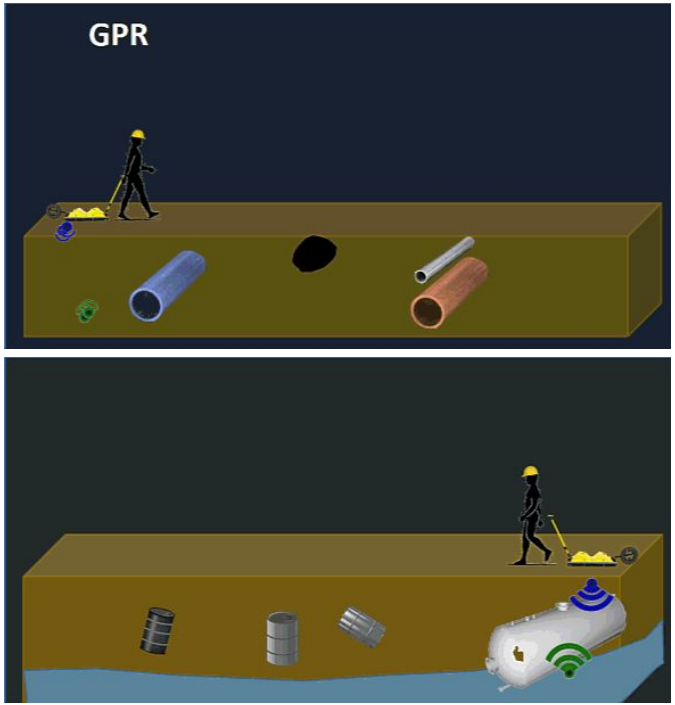
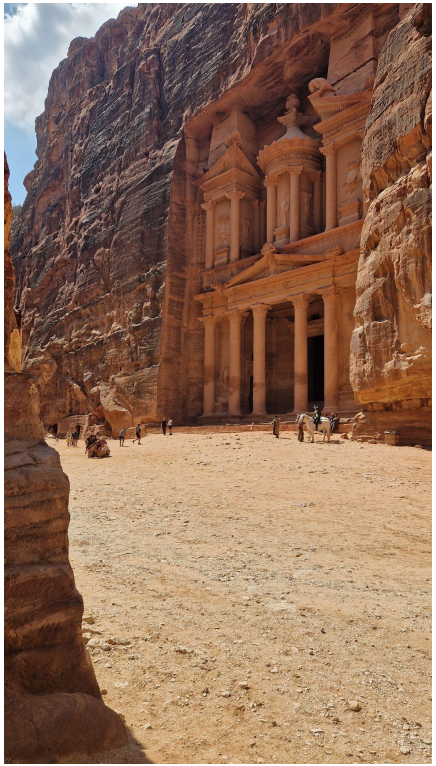


Today class

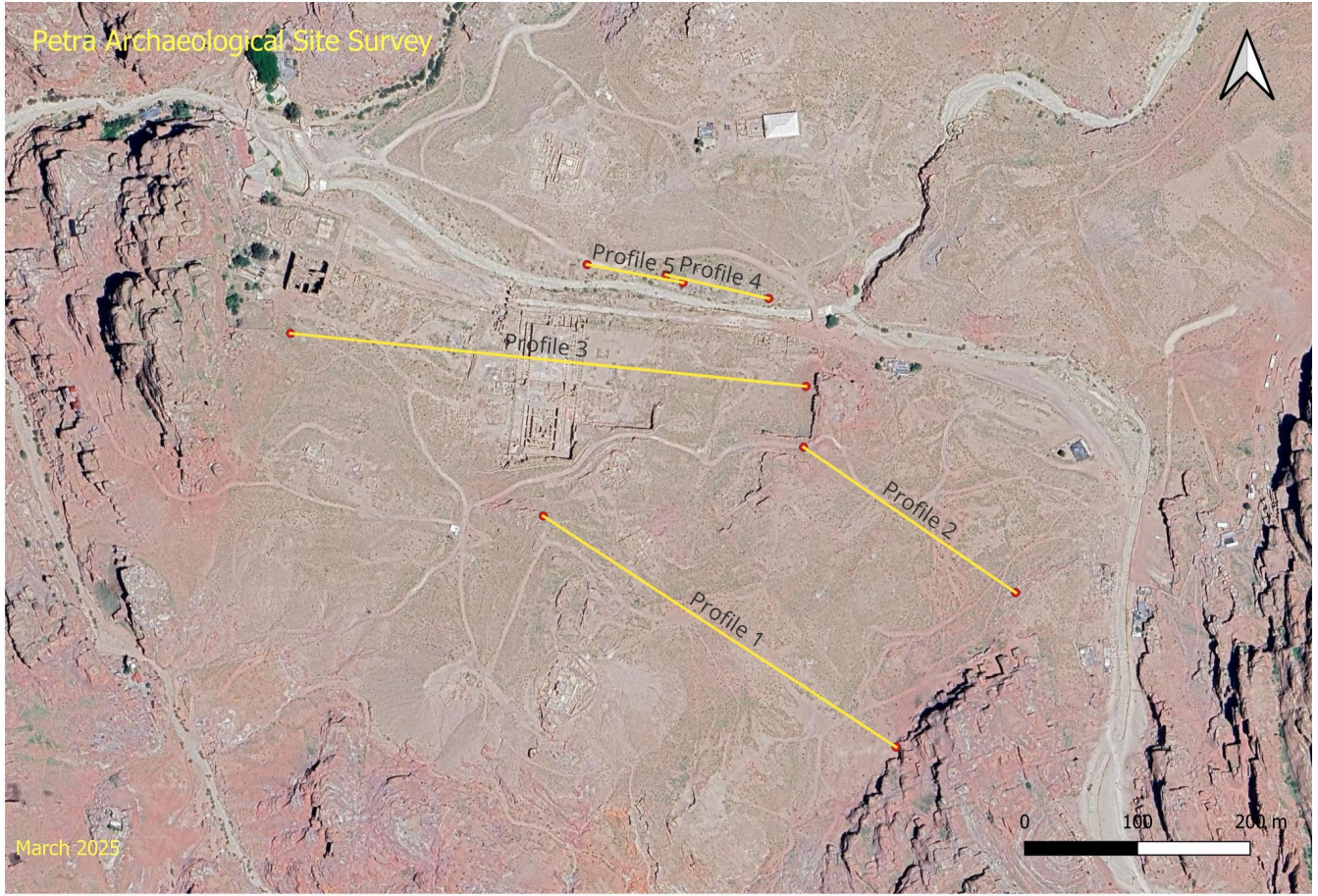
Real-world problems: Practical applications of geophysical methods

Case studies from the Chair

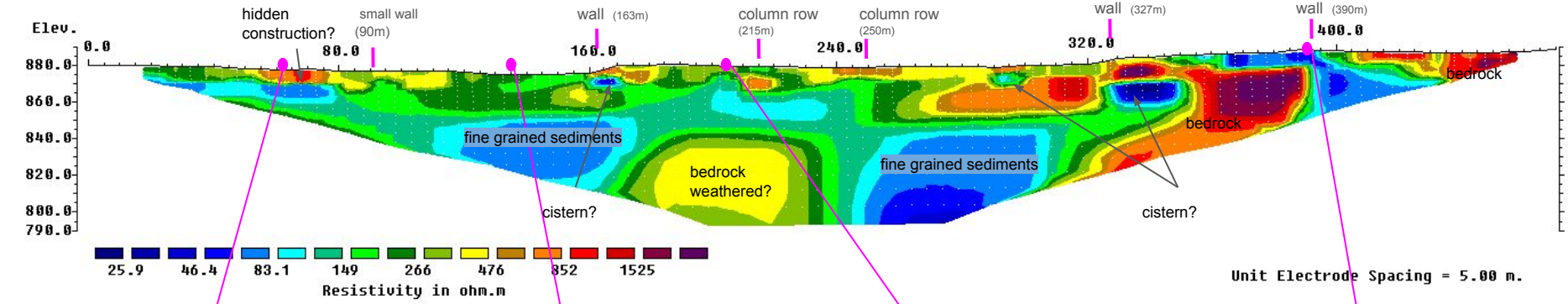
Geophysical ERT Survey in Petra – March 2025



Current Projects of the Chair - Petra Project



Profile 3



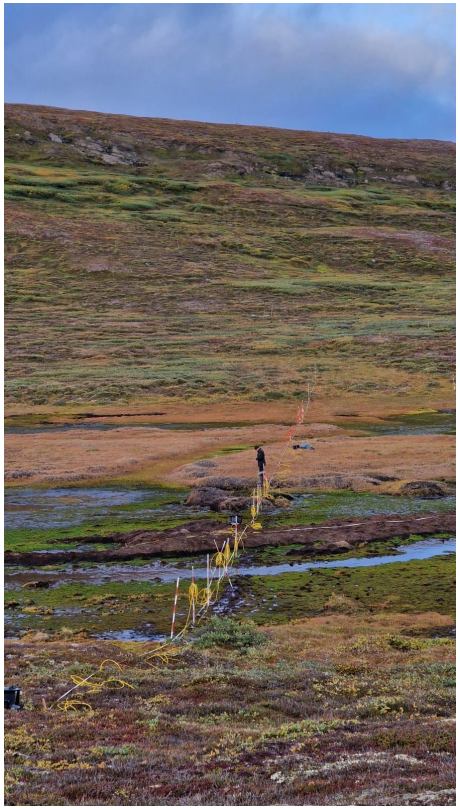
forward view at 60 m

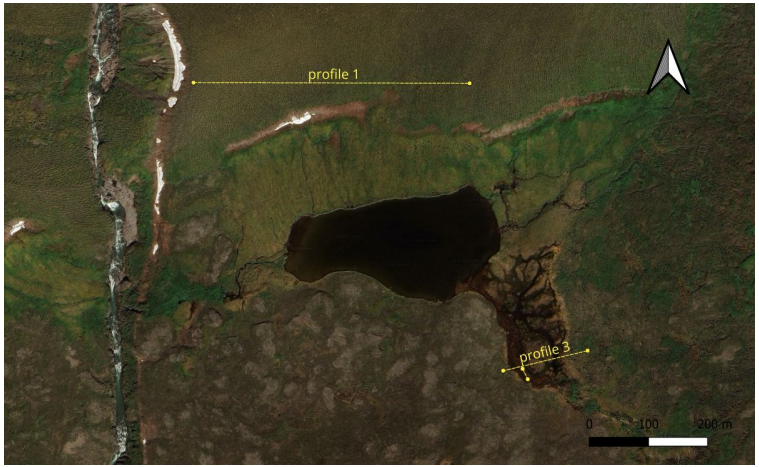
forward view at 135 m

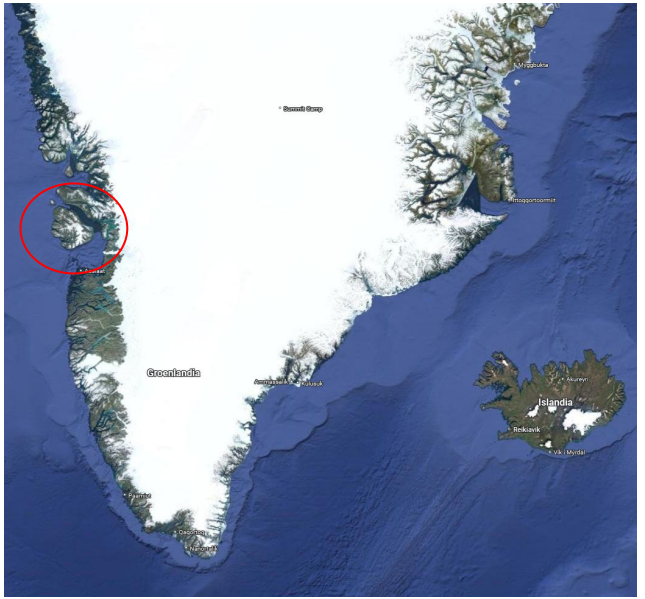
forward view at 203 m

backward view at 390 m

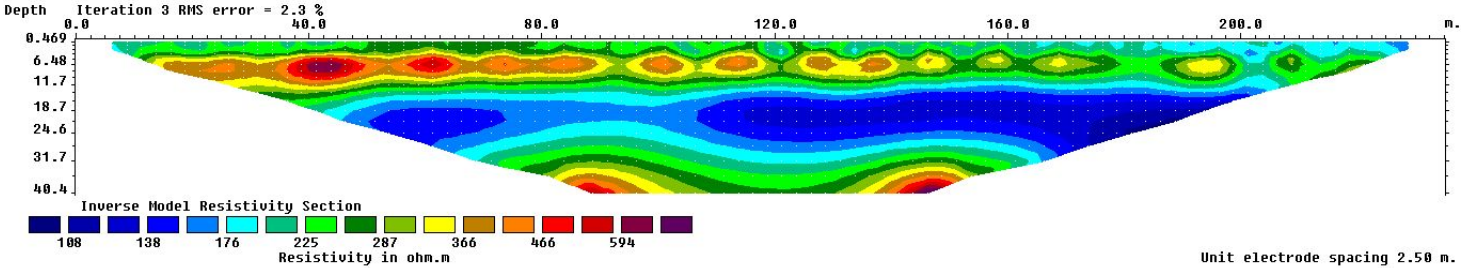
Geophysical ERT Survey Qeqertarsuaq - August 2025



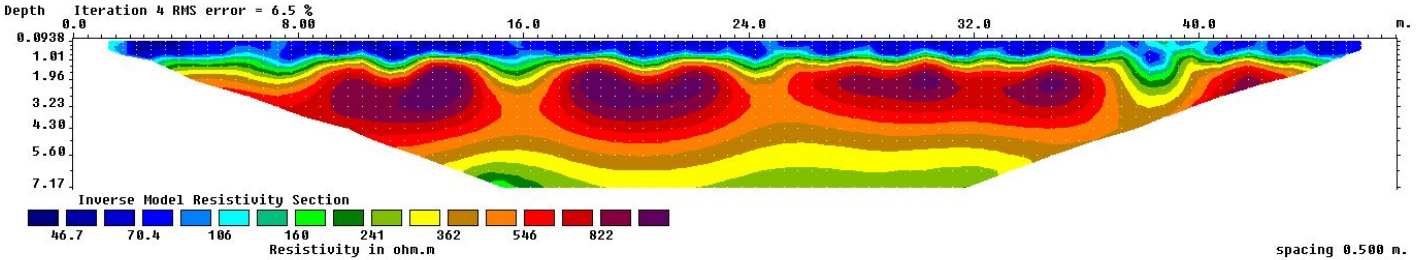




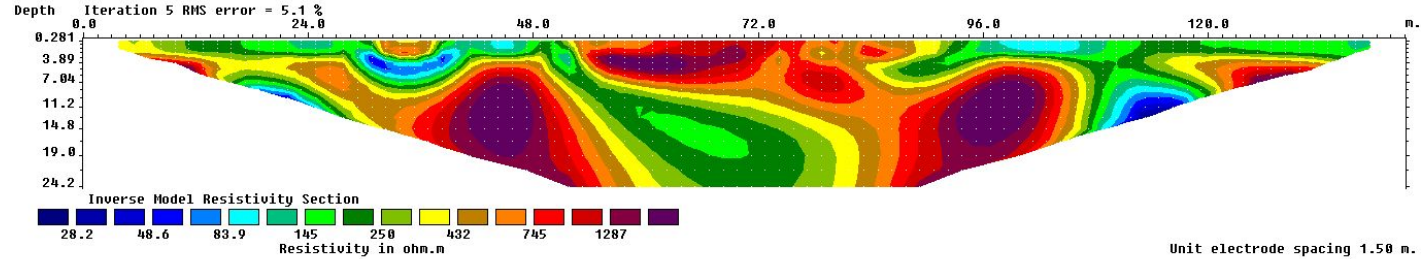
Preliminary results



Profile 1. 5m electrode distance



Profile 1. 1m electrode distance



Profile 3. 3m electrode distance

Brief overview of selected projects and case studies

BRINE

ERT, EM, MT, VLF → saltwater migration & storage potential

RoMo

Radiometric, Magnetometric → Development of the raw material sector of Mongolia with german know and how technology

Lebanon

ERT → Archaeological studies: exploration and 3D modeling of the geological subsurface of a Roman temple complex

Iran

ERT → Detection of mining cavities in a salt mine

Austria

ERT → Investigate prehistoric copper mining depressions along a main ore vein

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Exploration of weakness zones and saltwater upflow regions using geoelectrical and electromagnetic methods in the vicinity of a potential CO_2 storage site – Project brine

b-tu Brandenburg University of Technology Cottbus

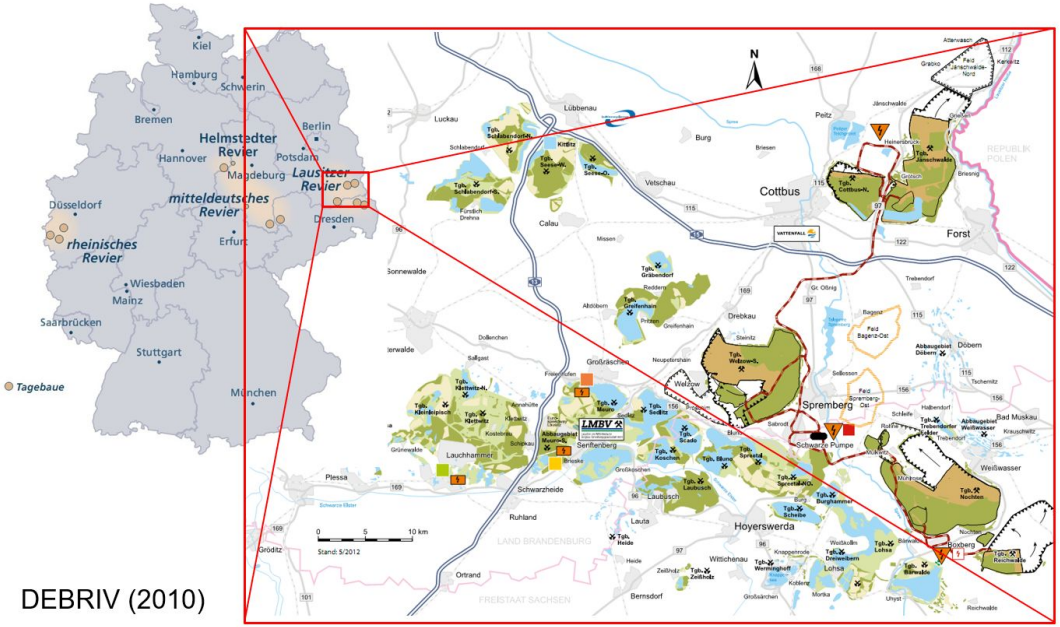
GFZ Helmholtz Centre Potsdam

GEOTECHNOLOGIEN

Federal Ministry of Education and Research

brine

Lignite mining in Germany



DEBRIV (2010)

Brandenburg University of Technology Cottbus

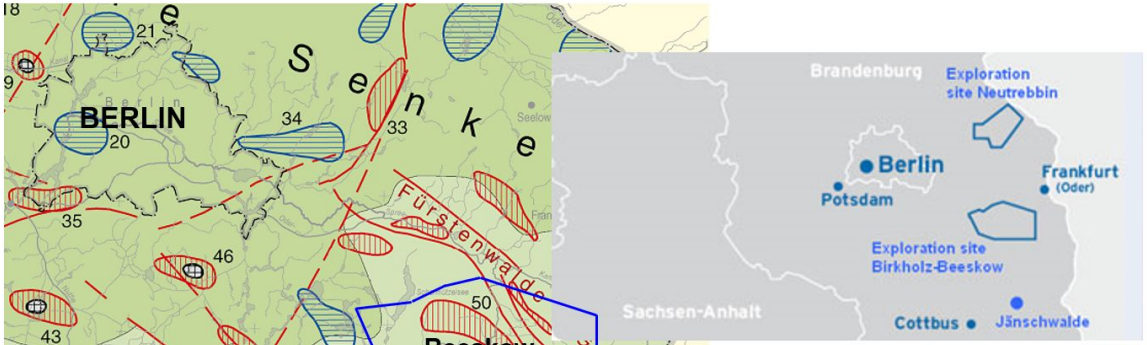
GFZ Helmholtz Centre Potsdam

GEOTECHNOLOGIEN

Federal Ministry of Education and Research

brine

Potential CO_2-Storage site: Salzstruktur Beeskow / Birkholz



Vattenfall (2010)

LBGR (2004)

Project brine:

CO_2-Storage in Eastern Brandenburg: Implications for a synergistic geothermal energy production and the design of an early warning system against groundwater salinization

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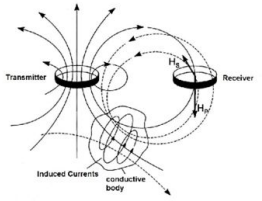
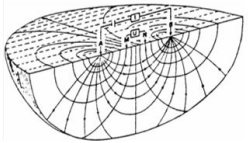
GEOTECHNOLOGIEN
Federal Ministry of Education and Research
brine

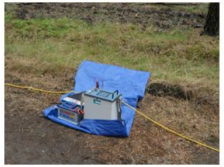
Sub-regional Investigation level Electromagnetic (EM)



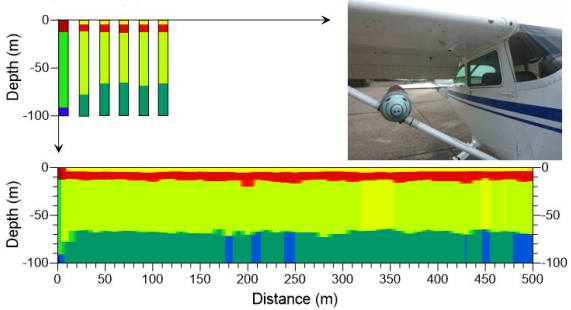
EM

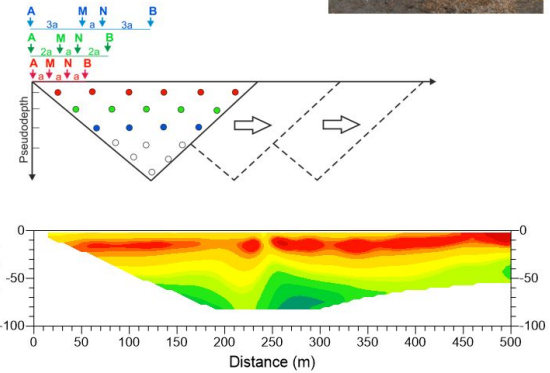


ERT

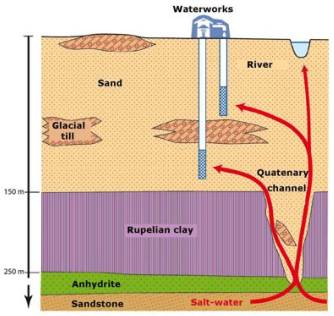


Airborne-VLF

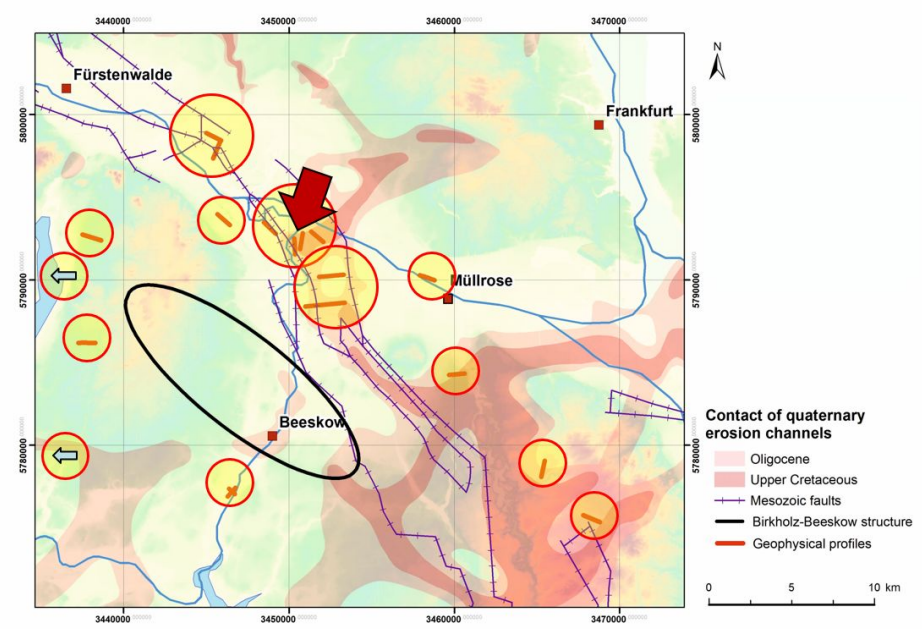




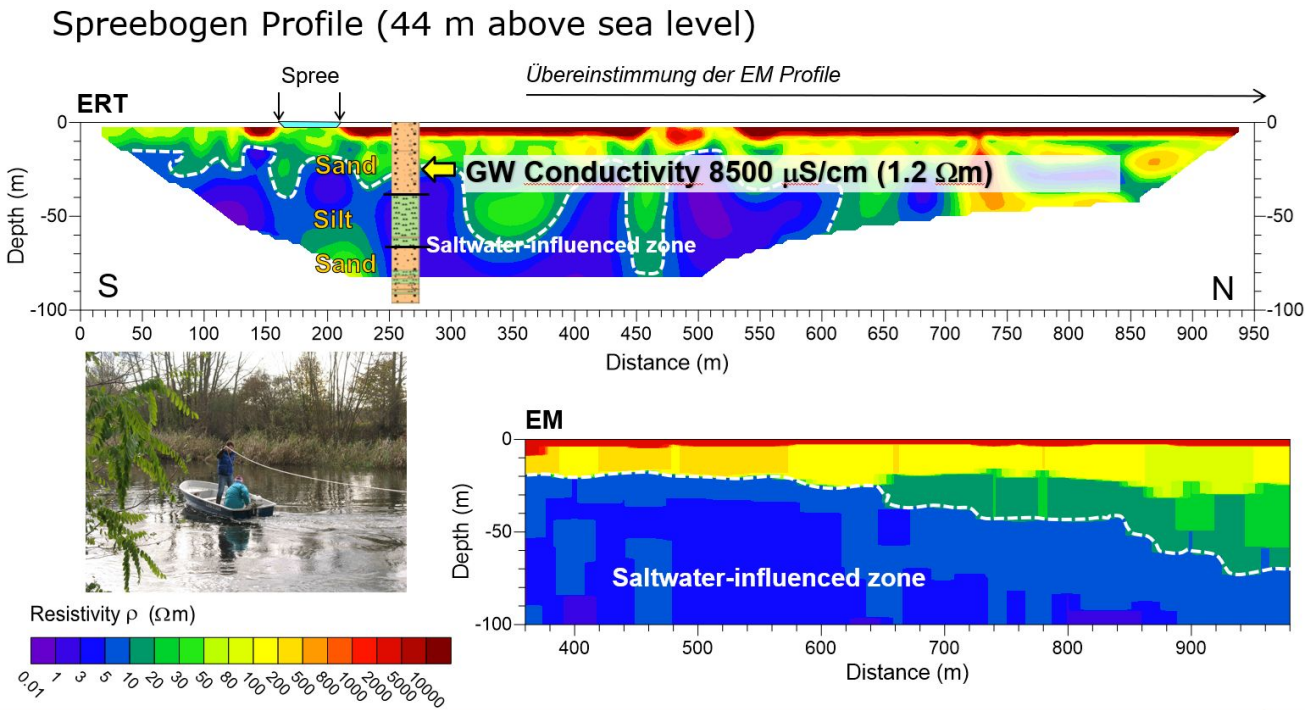
Preferred investigation areas: Quaternary channels and fault zones



Hermsdorf (2006)

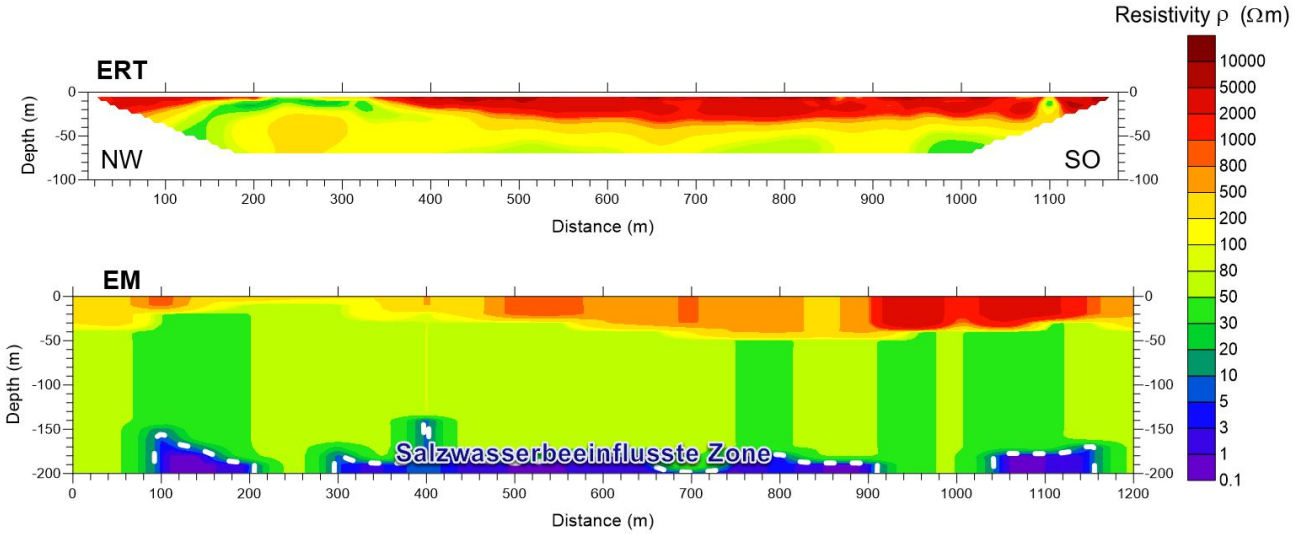


Results of the geophysical investigations

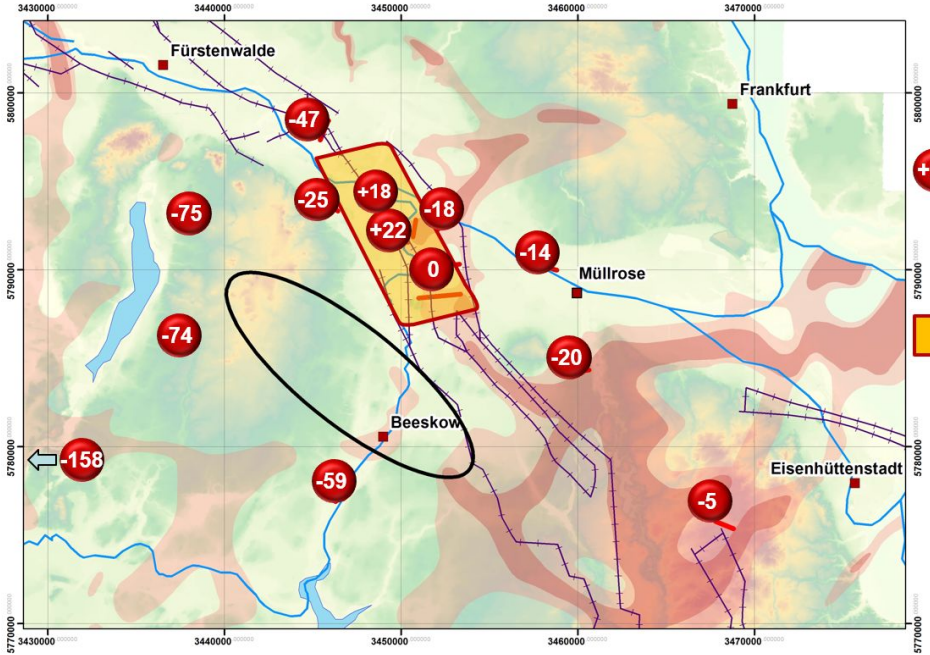


Results of the geophysical investigations

Profile Annenhof (75 m above sea level)



Depth of the saltwater – influenced zone(SWIZ)



+22 SWIZ (<10 Ωm, >1000µS/cm) in Meter above sea level

PVMZ (Preferential Vertical Migration Zone)

Brandenburg University of Technology Cottbus

GFZ Helmholtz Centre Potsdam

GEOTECHNOLOGIEN

Federal Ministry of Education and Research

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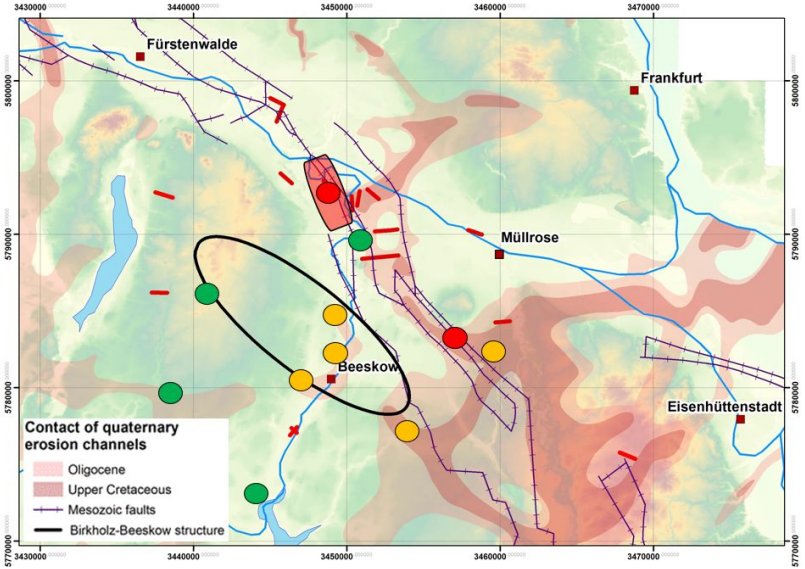
Preliminary concept of a geophysical early warning system against groundwater salinization based on MT, ERT und EM



Monitoring of CO_2 dispersion:

- MT at the surface
- CSMT transmitter / ERT in boreholes

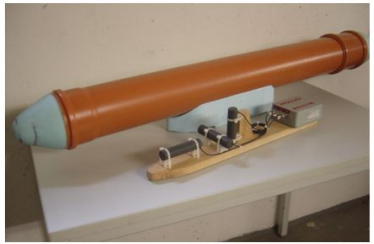
Monitoring the saltwater migration/ changes in the SWIZ:

- EM at the surface
- ERT in shallow monitoring boreholes

Local resistance to CCS (Carbon Capture and Storage)

Thank you for your attention!

Airborne VLF-EM investigations for detection of vertical saltwater migration



GFZ Helmholtz Centre Potsdam
btu Brandenburg University of Technology Cottbus
GEOTECHNOLOGIEN
Federal Ministry of Education and Research
brine

Brief overview of selected projects and case studies

BRINE

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ERT → Archaeological studies: exploration and 3D modeling of the geological subsurface of a Roman temple complex

Iran

ERT → Detection of mining cavities in a salt mine

Austria

ERT → Investigate prehistoric copper mining depressions along a main ore vein

Research Project RoMo - Raw Materials Mongolia

Background:

Agreement of Raw Material Partnership between Germany and Mongolia (signed 2011 by Angela Merkel and Sukhbaatar Batbold)

Aim:

- Further development of the raw material sector of Mongolia with german know how and technology
- Raw material supply for Germany in the long term

Way (beside others):

- Foundation of the GMIT (German-Mongolian Institute of Resources and Technology)
- Capacity building of Human Resources for the raw material sector at GMIT

Research Project RoMo - Raw Materials Mongolia

Targets of Project RoMo:

- Increasing research activities at GMIT in the raw material sector

Objectives:

- Performance of ground-based and **airborne investigations** for different raw material potentials (industrial minerals, metals, energy and ground water) in accordance with the **Mineral Resource and Petroleum Authority of Mongolia (MRPAM)**

Run-time: 01.01.2018 – 30.06.2019

Funding: BMZ via GIZ and DAAD

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Run-time: 01.01.2018 – 30.06.2019

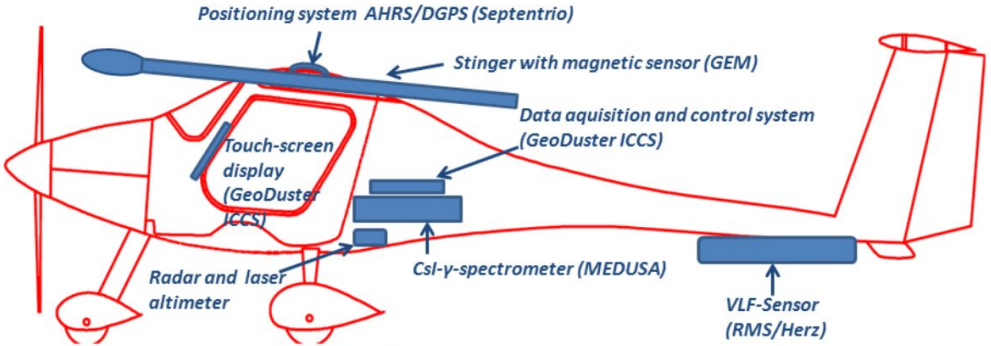
Funding: BMZ via GIZ and DAAD

Why Airborne Geophysics?

- Globally increasing demand for the raw materials, many underexplored regions – difficult to manage with ground-based methods
- Area-wide information is desired
- Prefereably contactless, “minimally invasive“ data acquisition.
- Increasing difficulty in obtaining land access permits
- Hardly any comercial systems availabe for small budgets
- Opportunity for teaching and research on current questions and new developments

Research Project RoMo - Raw Materials Mongolia

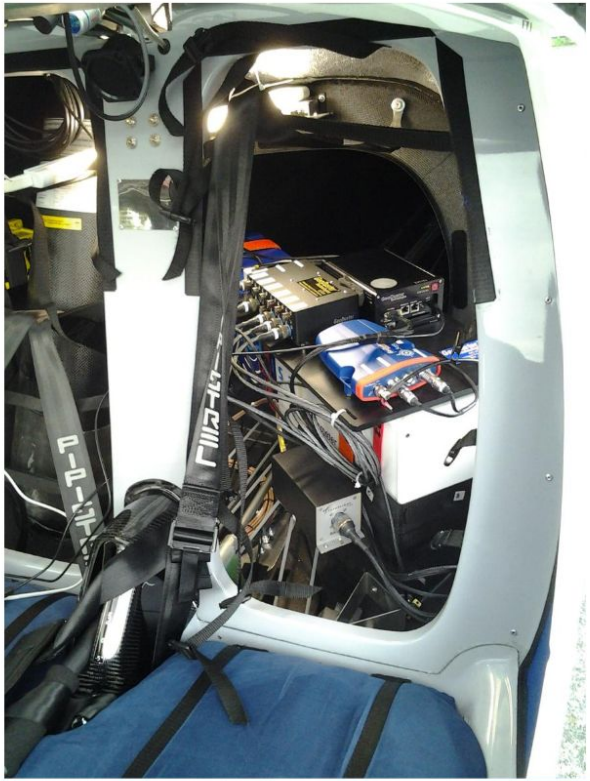
Tools: Airborne investigation system



Ausrüstung:

Blick ins Cockpit

Ausrüstung:



Blick in den Gepäckraum

Projekttagebuch

Rainer Herd

Die aerogeophysikalische Messplattform
D-MTUC

Brandenburg
University of Technology
Cottbus - Senftenberg

Projekttagebuch

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Die aerogeophysikalische Messplattform
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Research Project RoMo - Raw Materials Mongolia

Impressions

Research Project RoMo - Raw Materials Mongolia

Impressions

Projekttagebuch

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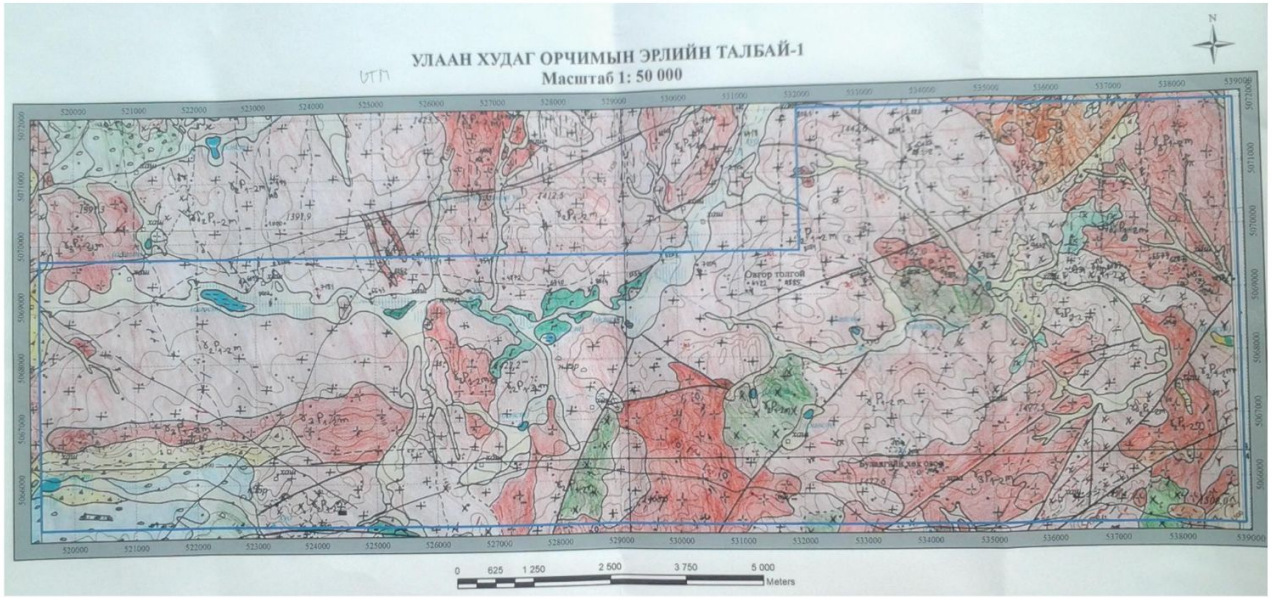
D-MTUC

Brandenburg
University of Technology
Cottbus - Senftenberg

Research Project RoMo - Raw Materials Mongolia

Impressions

Untersuchungsgebiet « Mandalgobi » (Distrikt Zentralgobi)



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Die aerogeophysikalische Messplattform
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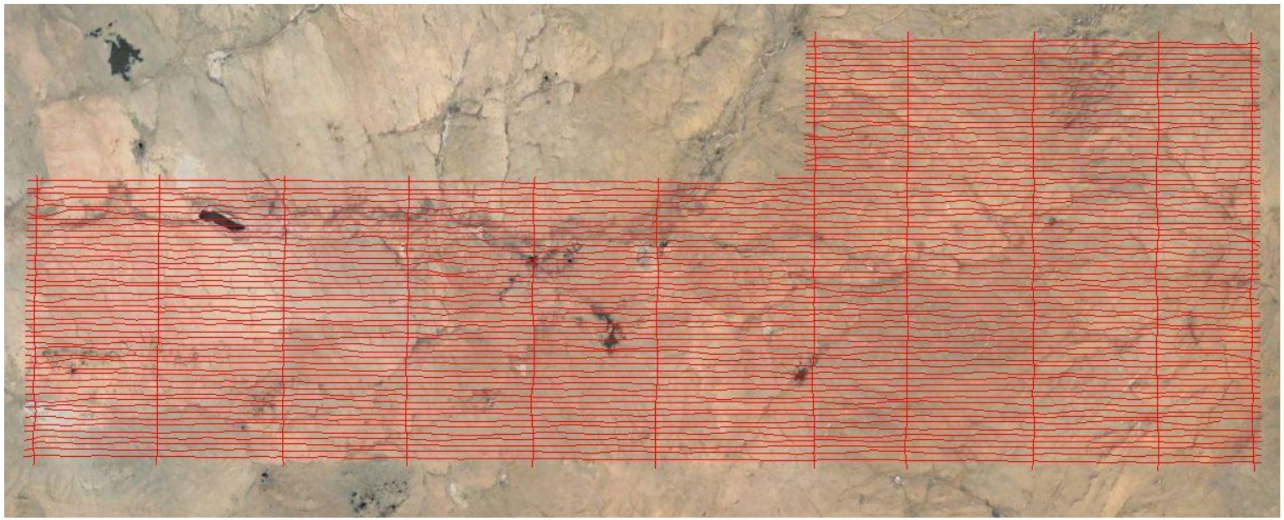
Brandenburg University of Technology Cottbus - Senftenberg

Untersuchungsgebiet « Mandalgobi » (Distrikt Zentralgobi))

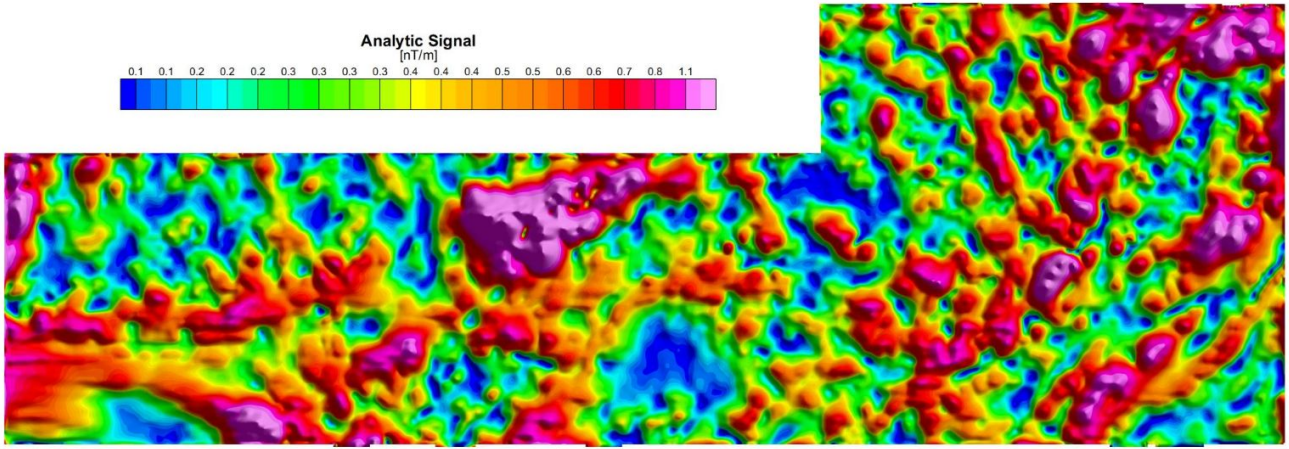


Areal extent: approx. 200 km²

Distance between lines: 100 m
Ground clearance: 100 m
Total: 2000 Line-km

Untersuchungsgebiet « Mandalgobi » (Distrikt Zentralgobi)

Ergebnisse: Magnetik



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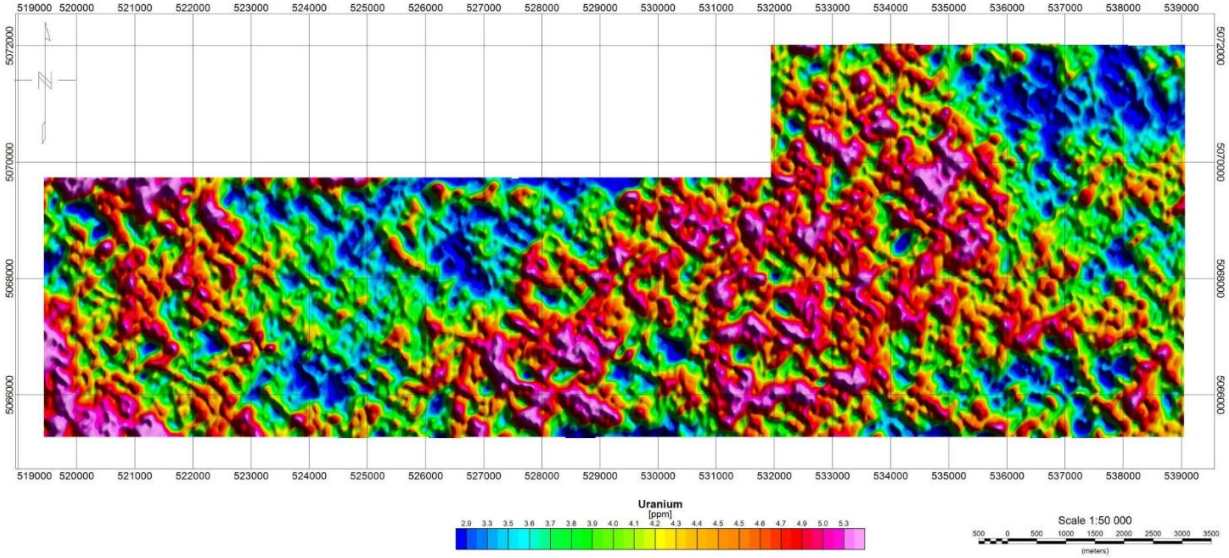
University of Technology
Cottbus - Senftenberg

Untersuchungsgebiet « Mandalgobi » (Distrikt Zentralgobi)

Ergebnisse: Radiometrie (Uran ppm)



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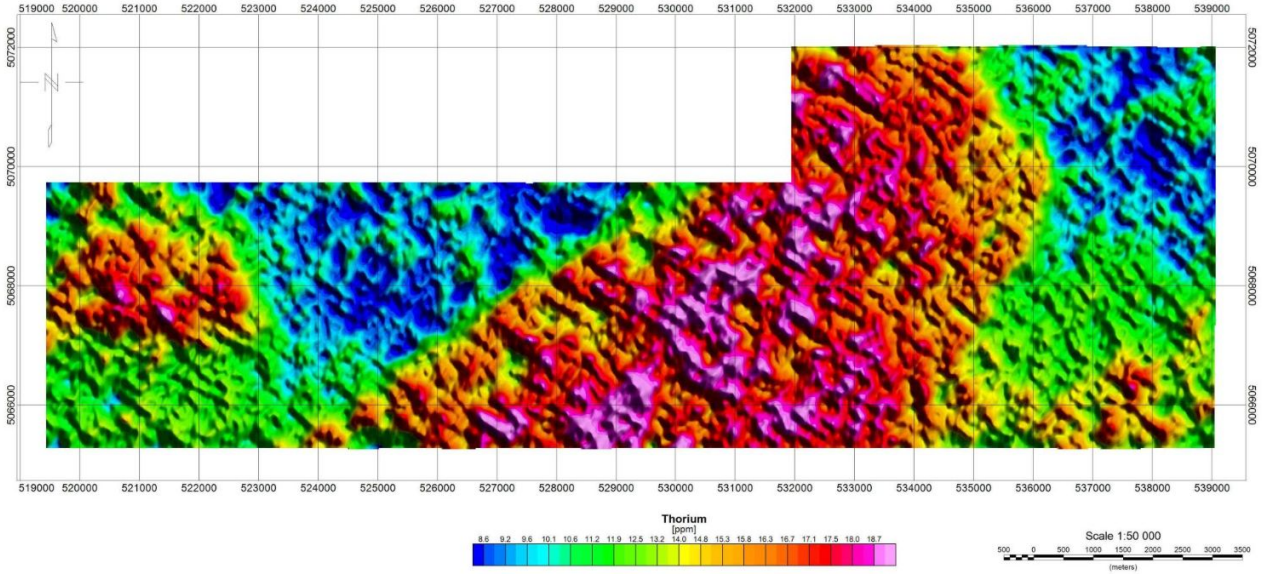
Untersuchungsgebiet « Mandalgobi » (Distrikt Zentralgobi)

Ergebnisse: Radiometrie (Thorium ppm)



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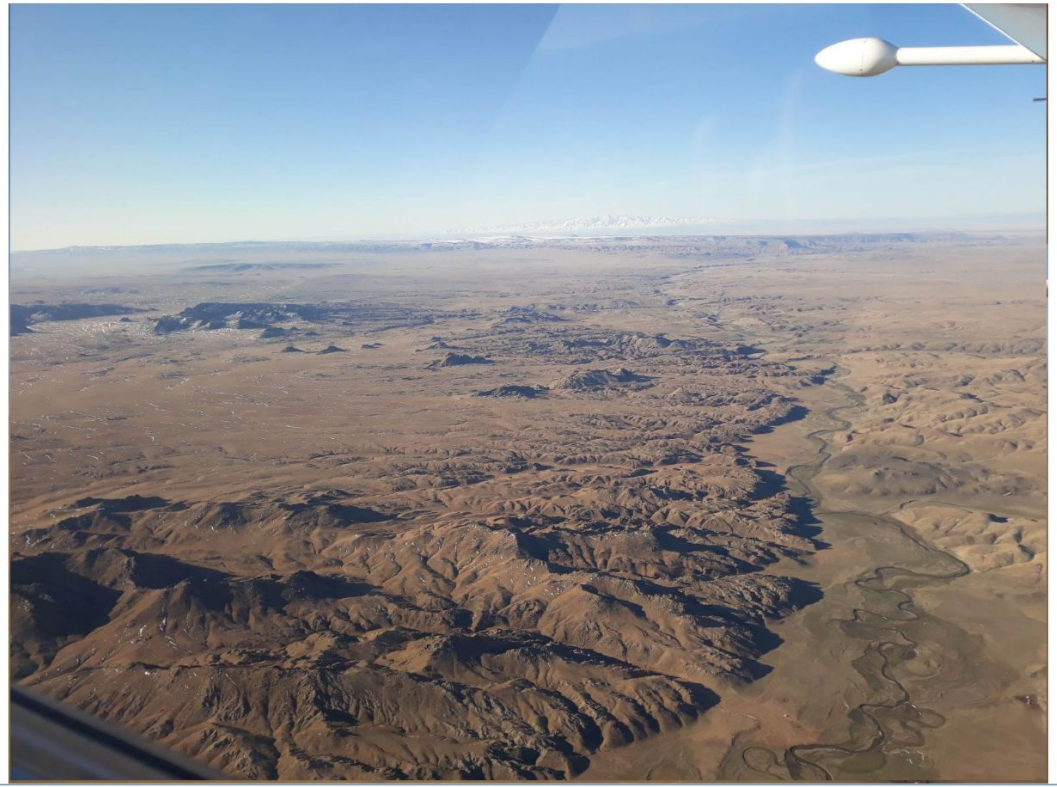
University of Technology
Cottbus - Senftenberg

Impressionen



Impressionen

Rainer Herd

Die aerogeophysikalische Messplattform
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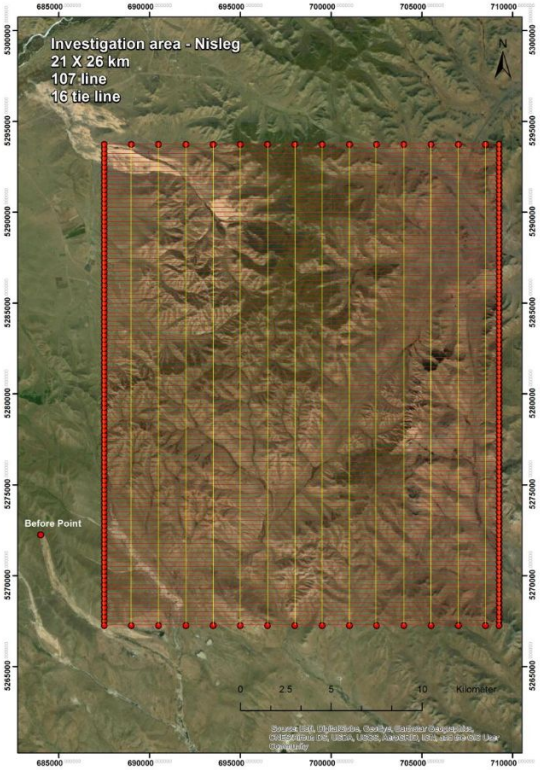
Cottbus - Senftenberg

The line grid of area « Janchiwlan »



Distance between lines: 250 m
Ground clearance: 100 m
Total: 2663 Line-km
Areal extent: 546 km²

GERMAN - MONGOLIAN INSTITUTE FOR RESOURCES AND TECHNOLOGY

Airborne geophysical investigations with an ultra-light aircraft in Mongolia (Phase 2022)

Brandenburg University of Technology Cottbus - Senftenberg

Brief overview of selected projects and case studies

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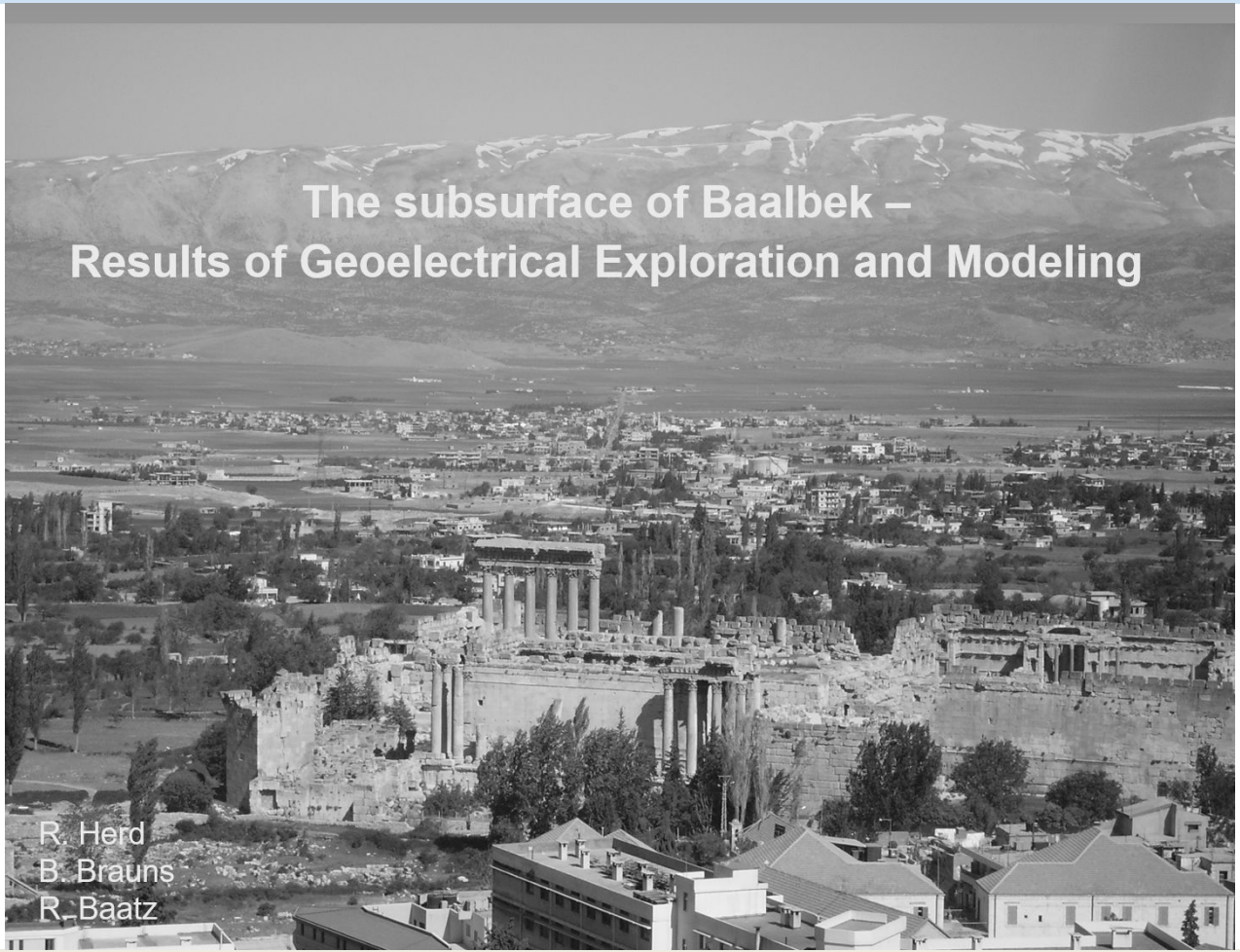
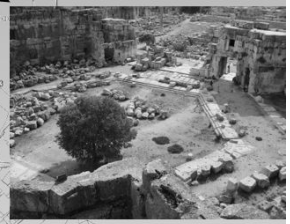
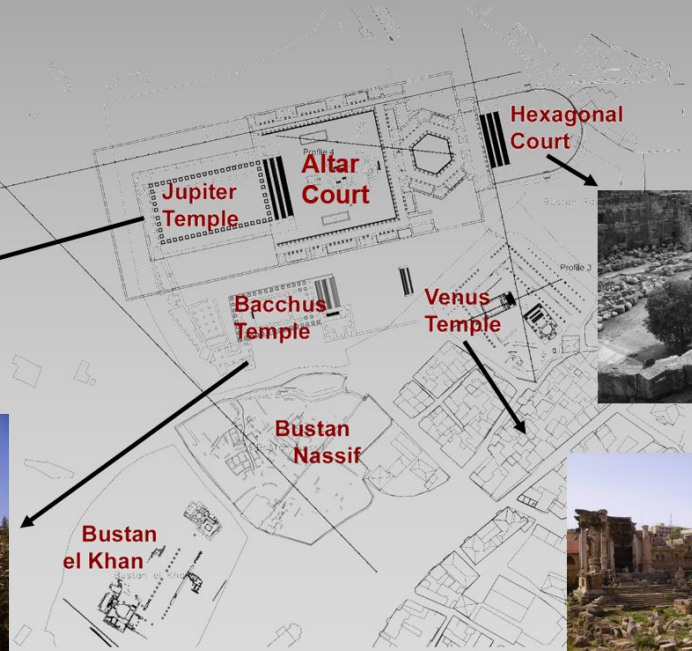

The subsurface of Baalbek –
Results of Geoelectrical Exploration and Modeling
R. Herd
B. Brauns
R. Baatz

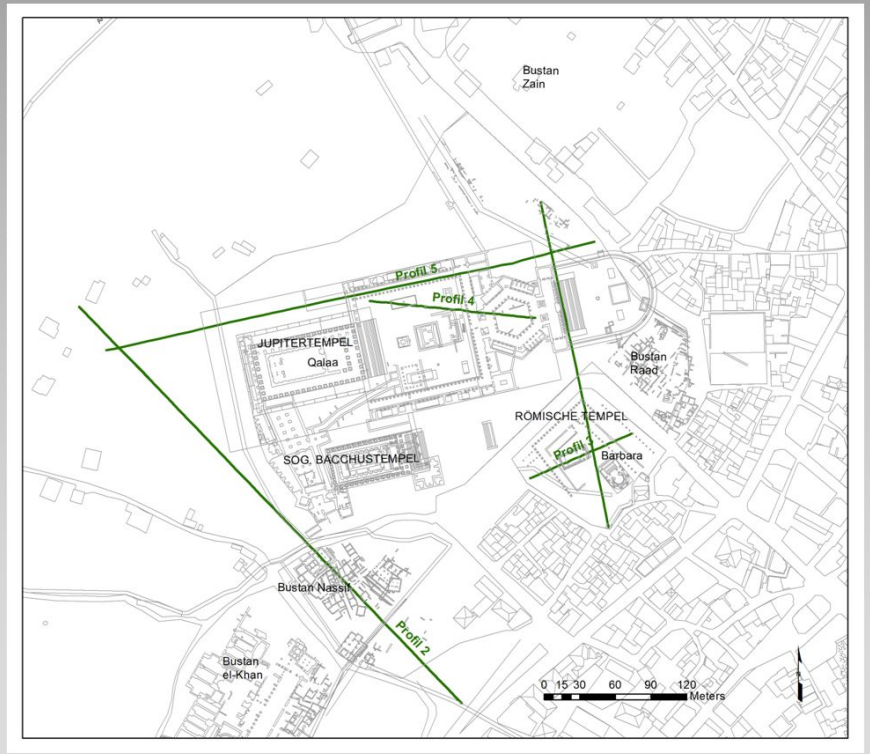
Investigation of Geological Structures





Der Untergrund von Baalbek –
Ergebnisse geoelektrischer Erkundung und Modellierung

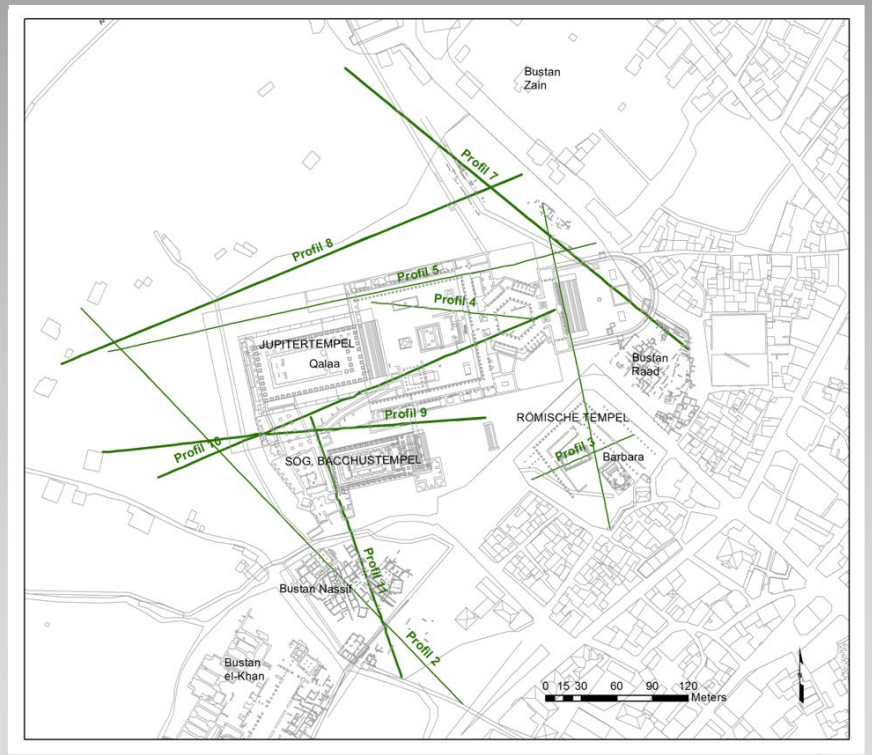
Areas of investigation 2008



Der Untergrund von Baalbek –
Ergebnisse geoelektrischer Erkundung und Modellierung

b-tu Brandenburgische Technische Universität Cottbus

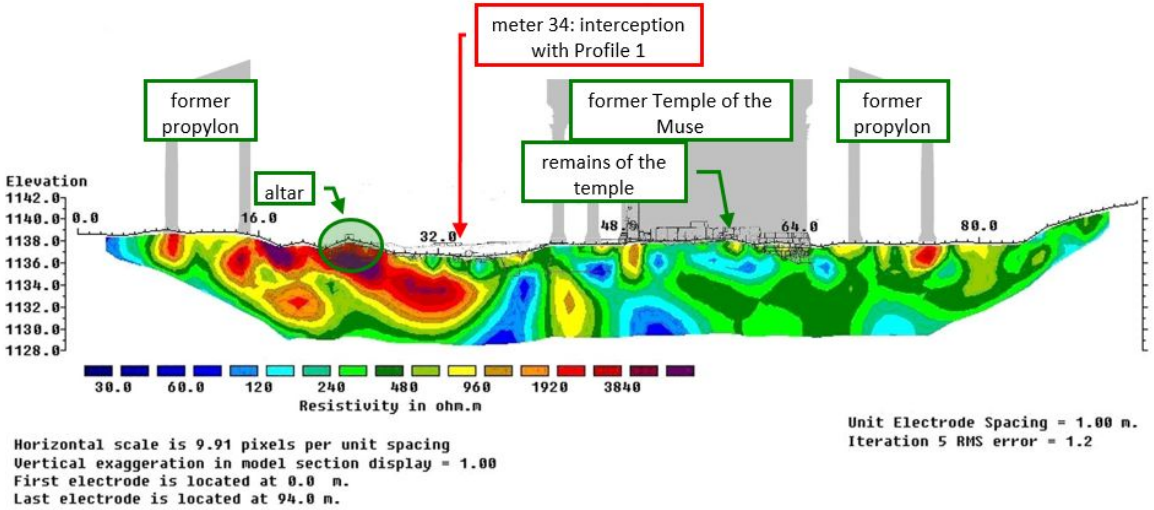
Areas of investigation 2011



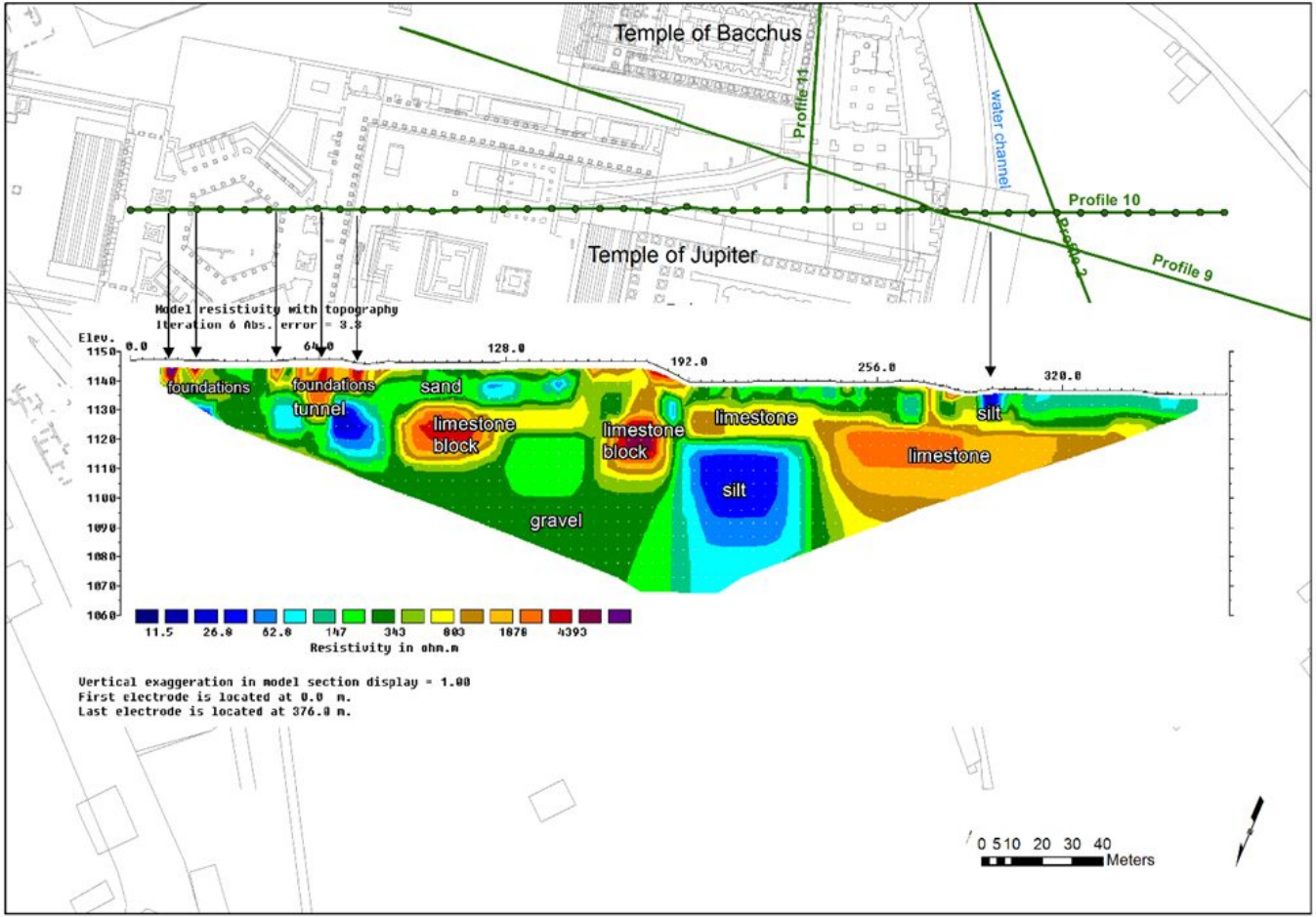
Der Untergrund von Baalbek –
Ergebnisse geoelektrischer Erkundung und Modellierung

b-tu Brandenburgische Technische Universität Cottbus

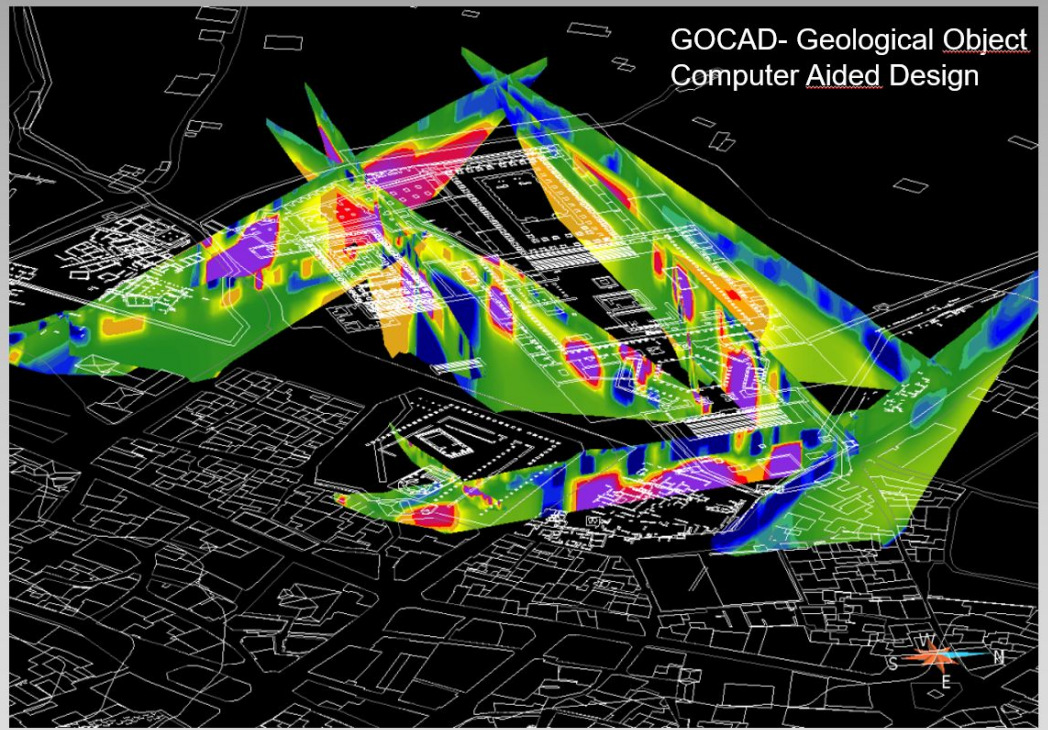
Previous Projects of the Chair - Lebanon



Previous Projects of the Chair - Lebanon



3D representation using GOCAD



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Ergebnisse geoelektrischer Erkundung und Modellierung

b-tu
Brandenburgische Technische Universität Cottbus

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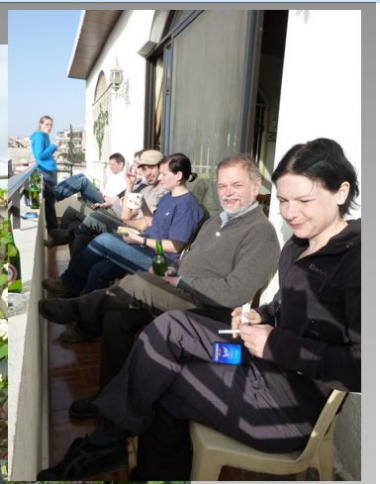

Herzlichen Dank für die Unterstützung
und angenehme Zusammenarbeit!

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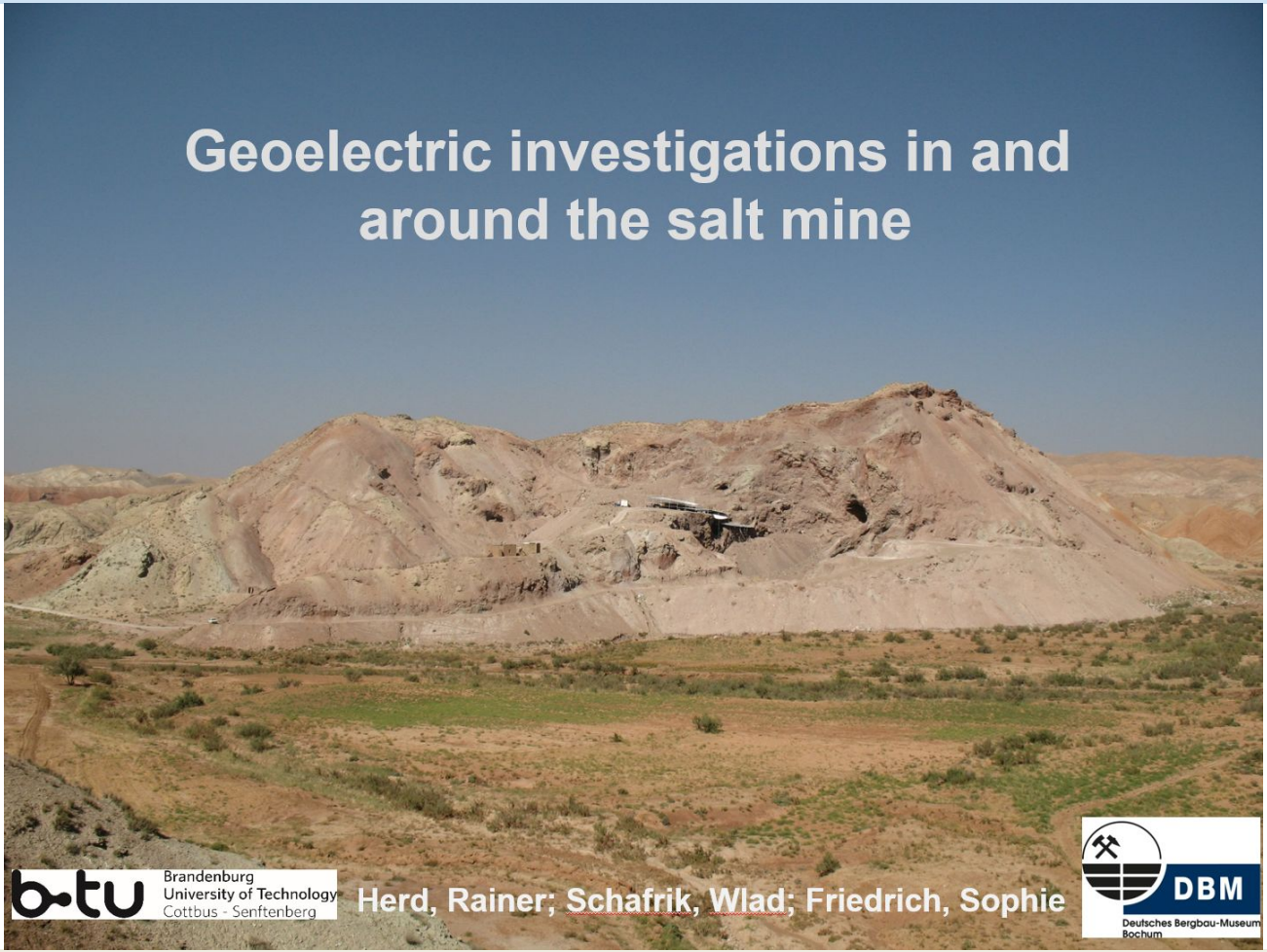
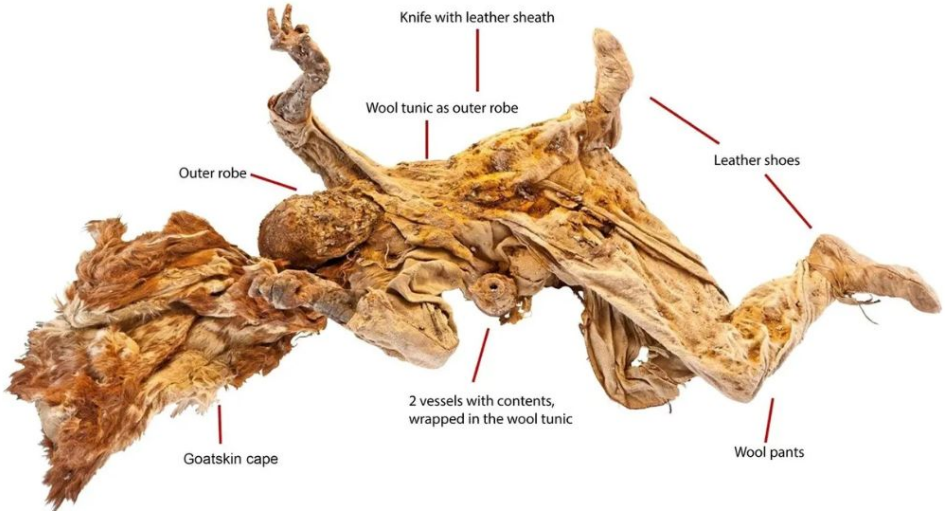
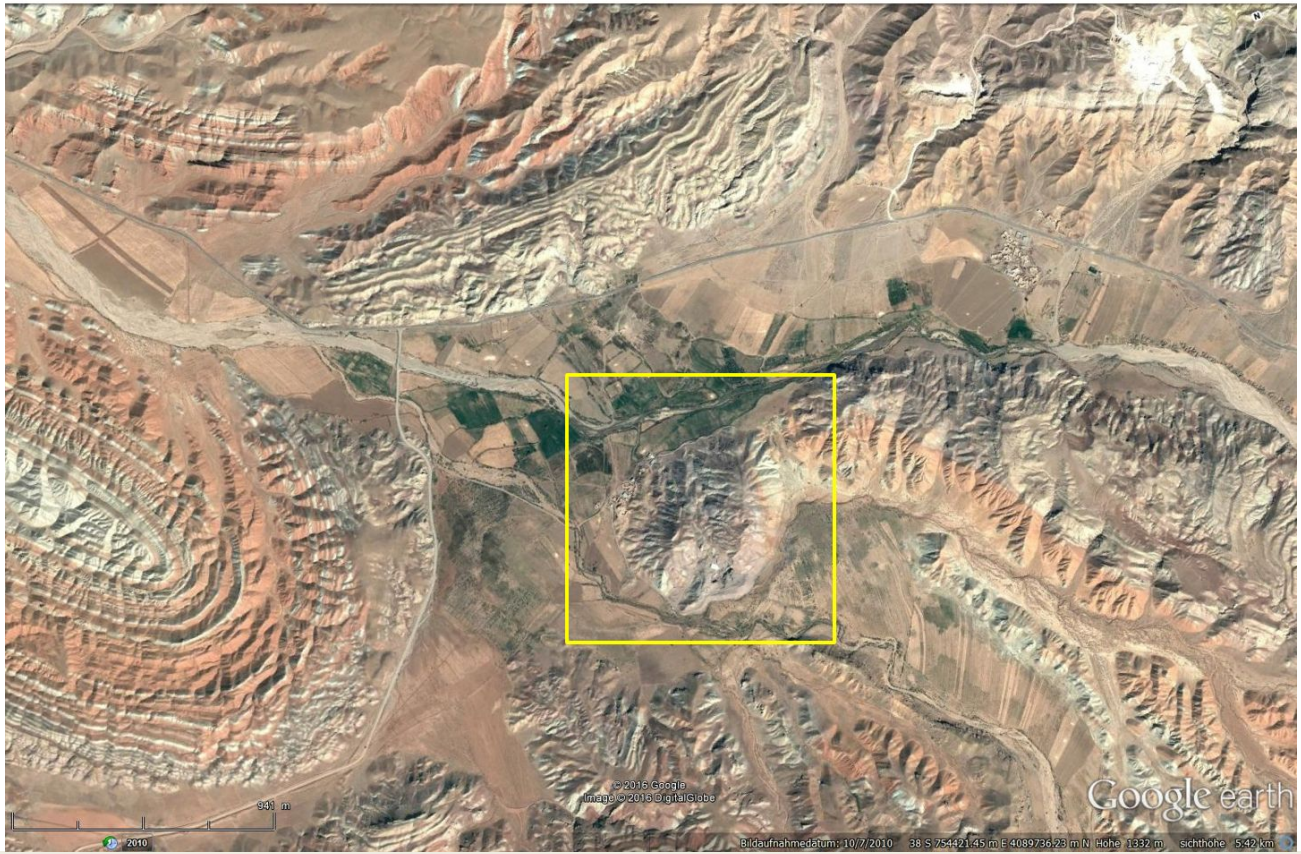

Geoelectric investigations in and around the salt mine
Brandenburg University of Technology Cottbus - Senftenberg
Herd, Rainer; Schafrik, Wlad; Friedrich, Sophie
DBM
Deutsches Bergbau-Museum Bochum

Saltman found

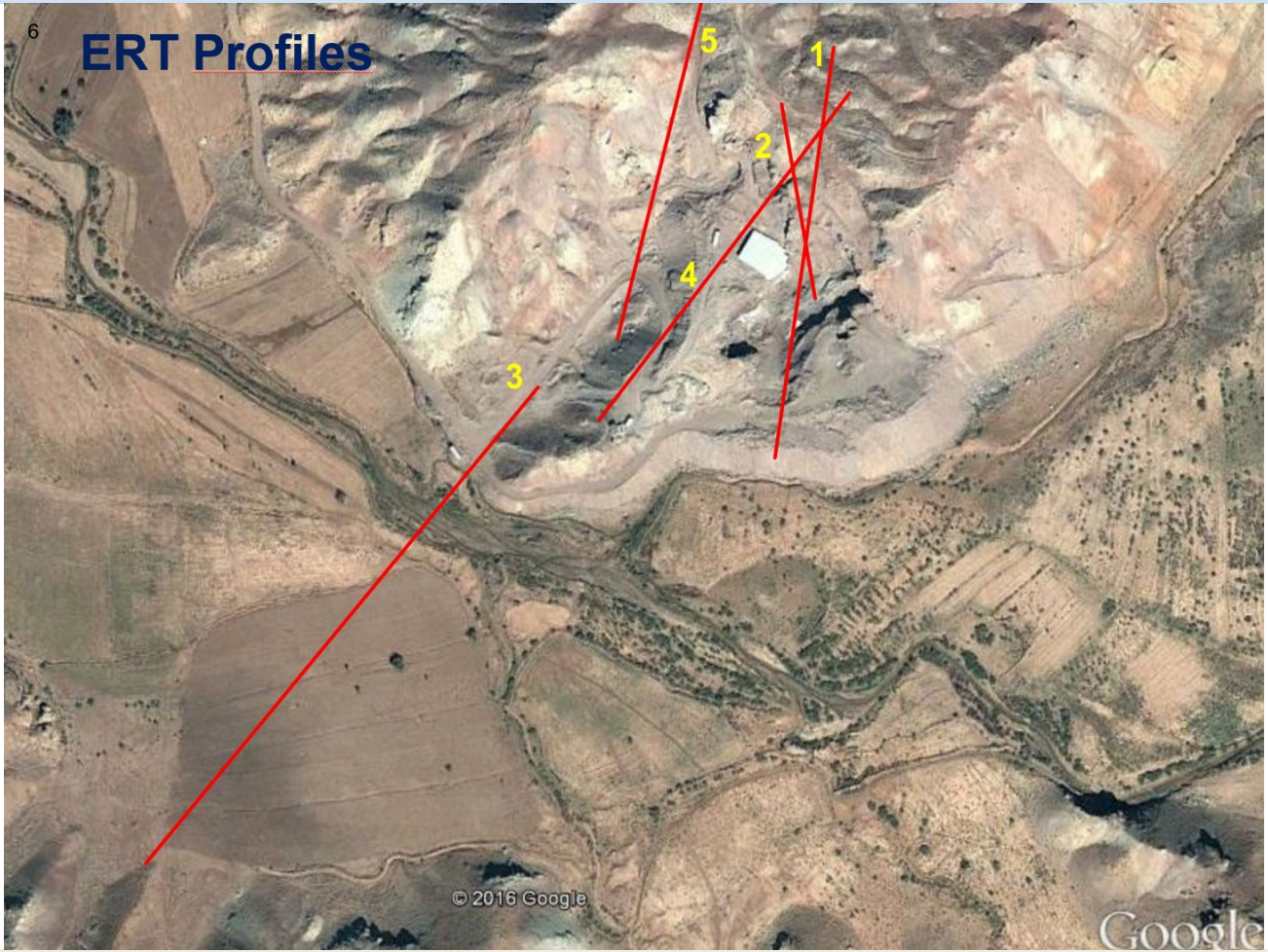


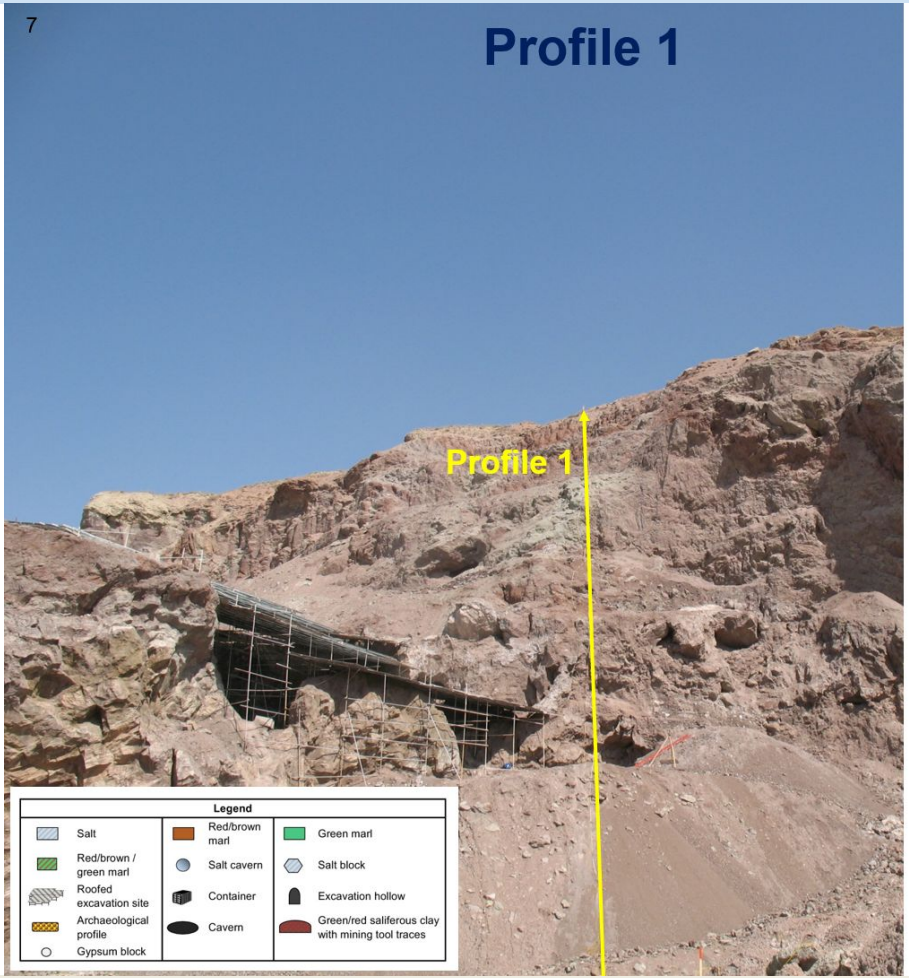
2

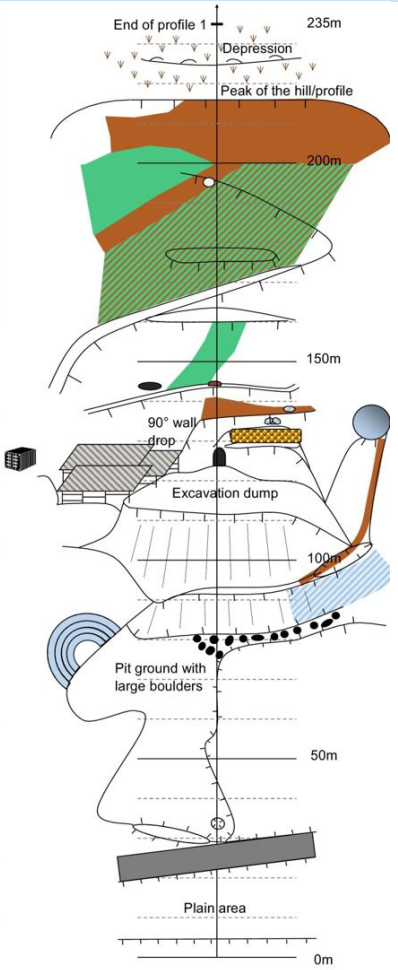
Chehrābād Douzlākh and its surrounding region



Salt mine Chehrābād Douzlākh

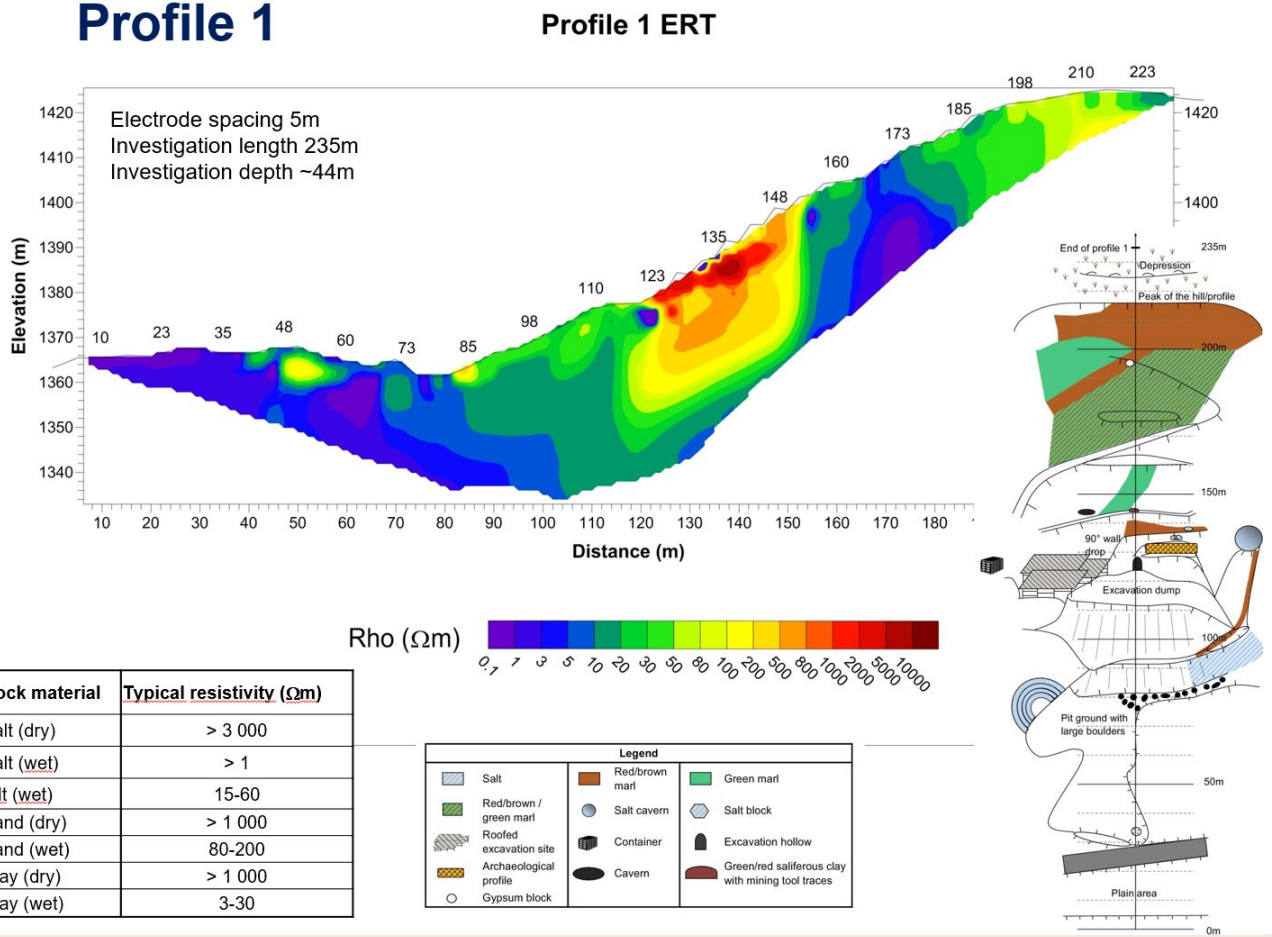

6
ERT Profiles
5
1
2
4
3
© 2016 Google
Google


7
Profile 1
Profile 1
Legend
Salt
Red/brown / green marl
Roofed excavation site
Archaeological profile
Gypsum block
Red/brown marl
Salt cavern
Container
Cavern
Green marl
Salt block
Excavation hollow
Green/red saliferous clay with mining tool traces


End of profile 1
235m
Depression
Peak of the hill/profile
200m
150m
90° wall drop
Excavation dump
100m
Pit ground with large boulders
50m
Plain area
0m

8

Profile 1



Rock material	Typical resistivity (Ωm)
Salt (dry)	> 3 000
Salt (wet)	> 1
Silt (wet)	15-60
Sand (dry)	> 1 000
Sand (wet)	80-200
Clay (dry)	> 1 000
Clay (wet)	3-30

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Figure 1: Research region of the HiMAT project in Austria.

Source: Geographic guide 1992.

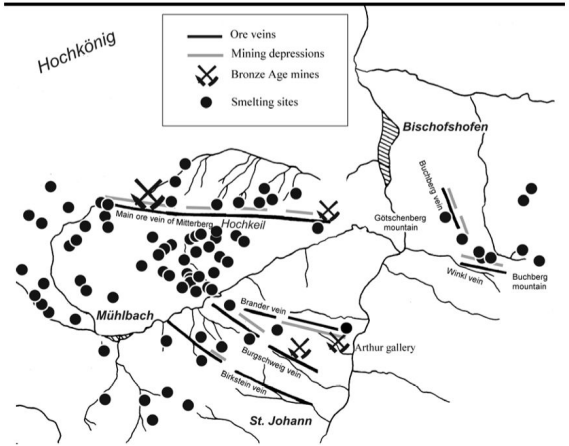
The Mitterberg region near Salzburg (I), Schwaz/ Brixlegg in Tirol (II) and the Bartholomäberg and Silbertal in the Montafon region (III). (HiMAT 2008)

Figure 2: The Nebra sky disk.

Source: Meller 2004.

Figure 3: Entrance to the Arthur gallery.

Figure 4: The mining sites in the Mitterberg region and connected smelting sites.



Source: Adapted from Stöllner.

Figure 5: One of the mining depressions in the Mitterberg region.

Figure 6: The Arthurhouse with the Hochkönig mountain range in the background.

Figure 7: The Mitterberg stream

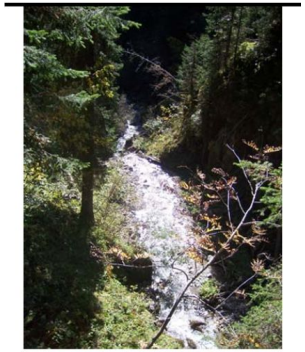


Figure 8: Troyground with the old peat deposit in the foreground.

Figure 13: The ore.

Distribution of the profiles on the northern flank of the Hochkail mountain range

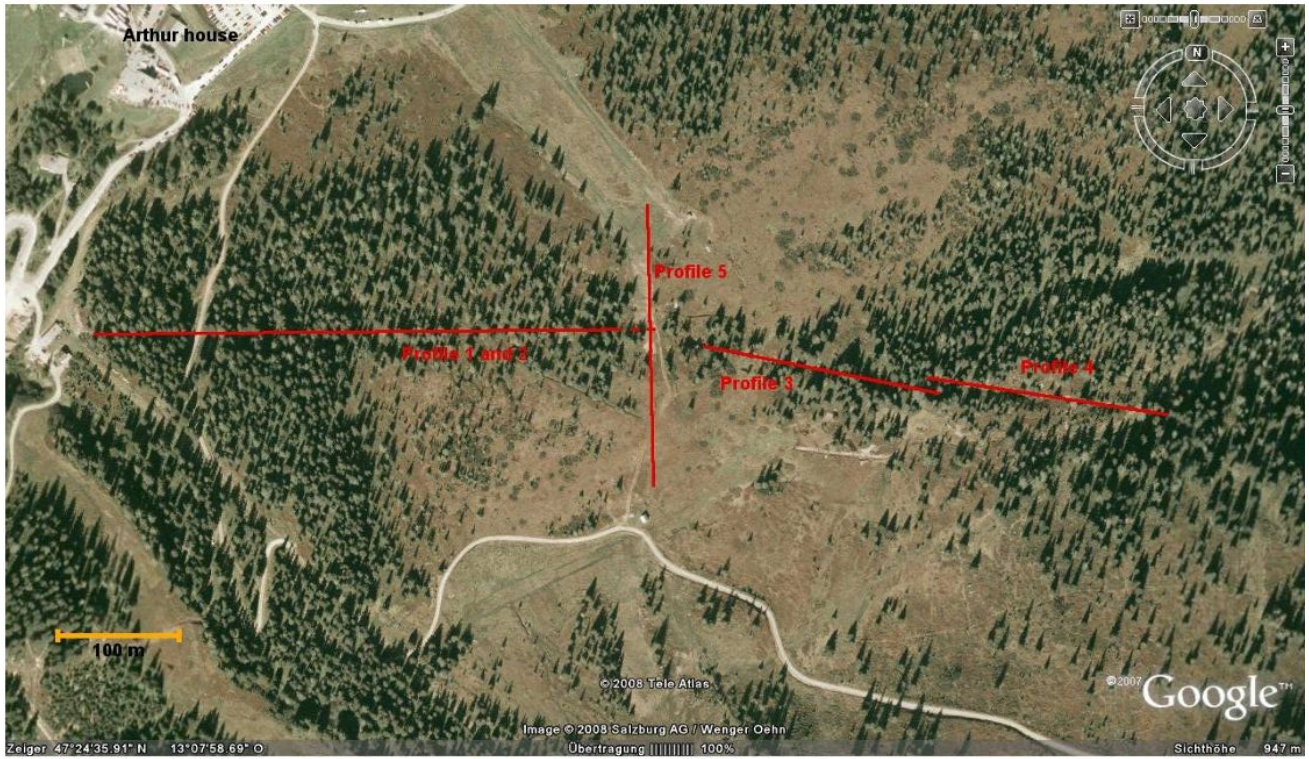


Figure 37: Start of the profile, view in direction of the skiing-lift valley station.

Photo: R.Herd.

Figure 38: The upmost plain area, view in direction of the profile begin.

Photo: R.Herd.

Figure 39: The water ditch and the fallen tree, view in direction of profile begin.

Photo: R.Herd.

Figure 40: The fallen tree on the profile line.

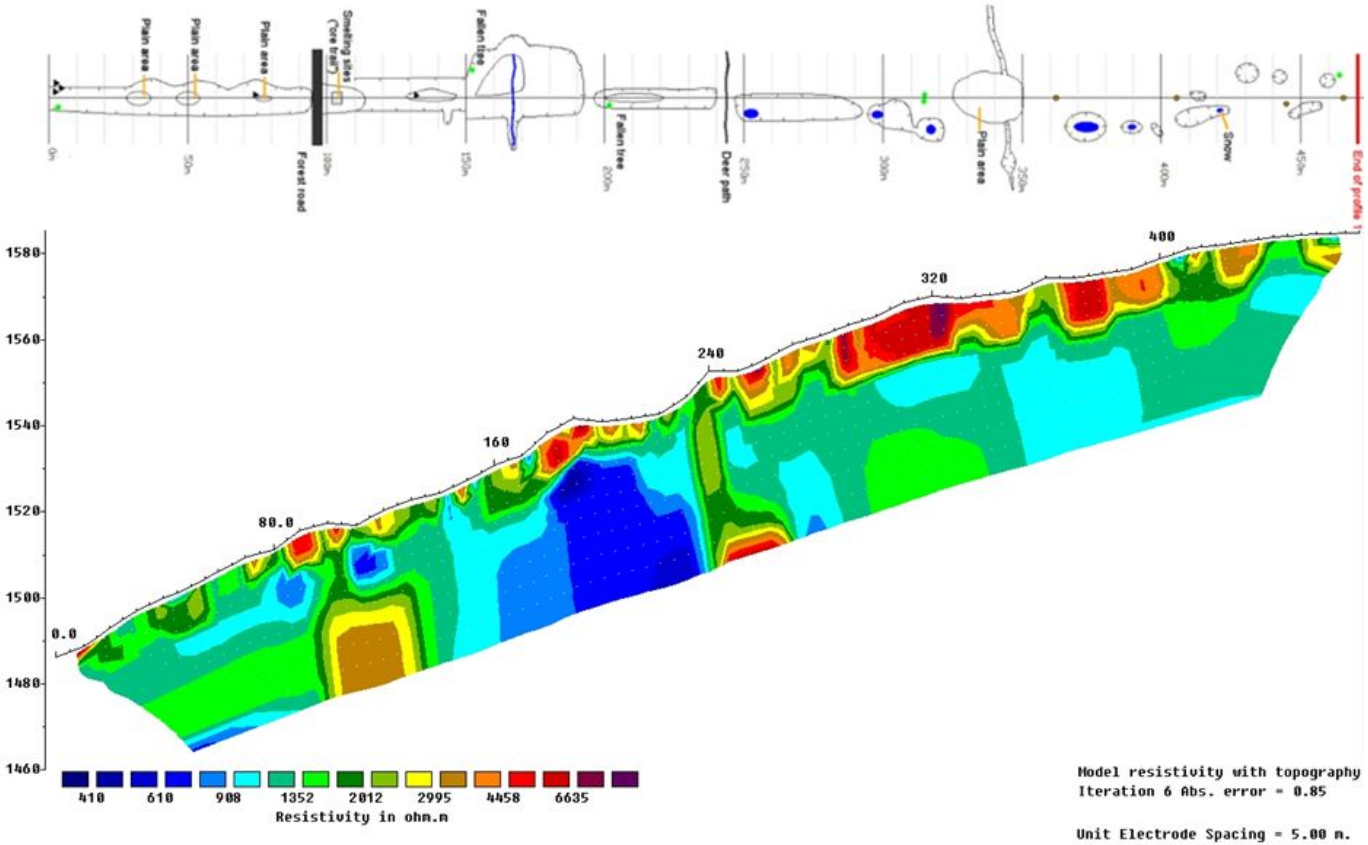
Photo: R.Herd.

Figure 41: The beginning of the troyground.

Photo: R.Herd.

Figure 42: The extension of profile 1 and 2 cutting the forestry path on the troyground.

Photo: R.Herd.


Plain area
Plain area
Plain area
Smelting sites (Core trail)
Fallen tree
Fallen tree
Deer path
Plain area
Snow
End of profile 1
0m
50m
100m
150m
200m
250m
300m
350m
400m
450m
Forest road
1580
1560
1540
1520
1500
1480
1460
0.0
80.0
160
240
320
400
410
610
908
1352
2012
2995
4458
6635
Resistivity in ohm.m
Model resistivity with topography
Iteration 6 Abs. error = 0.85
Unit Electrode Spacing = 5.00 m.